

Deep learning for image processing

STAT 4710

November 13, 2025

Where we are

- ✓ **Unit 1:** R for data mining
- ✓ **Unit 2:** Prediction fundamentals
- ✓ **Unit 3:** Regression-based methods
- ✓ **Unit 4:** Tree-based methods
- Unit 5:** Deep learning

Lecture 1: Deep learning preliminaries

Lecture 2: Training and optimization of NNs

Lecture 3: Deep learning for images

Lecture 4: Deep learning for text

Lecture 5: Unit review and quiz in class

Plan for Unit 5: concepts & hands-on

Lecture 1: Deep learning preliminaries

- Predictive models as graphs
- Training via optimization: gradient descent
- Hands-on: simple NN on MNIST in R torch

Lecture 2: Neural networks

- Multi-layer predictive models
- Stochastic gradient descent
- Hands-on: multi-layer NN for MNIST in R torch with newly learned training options

Lecture 3: Deep learning for images

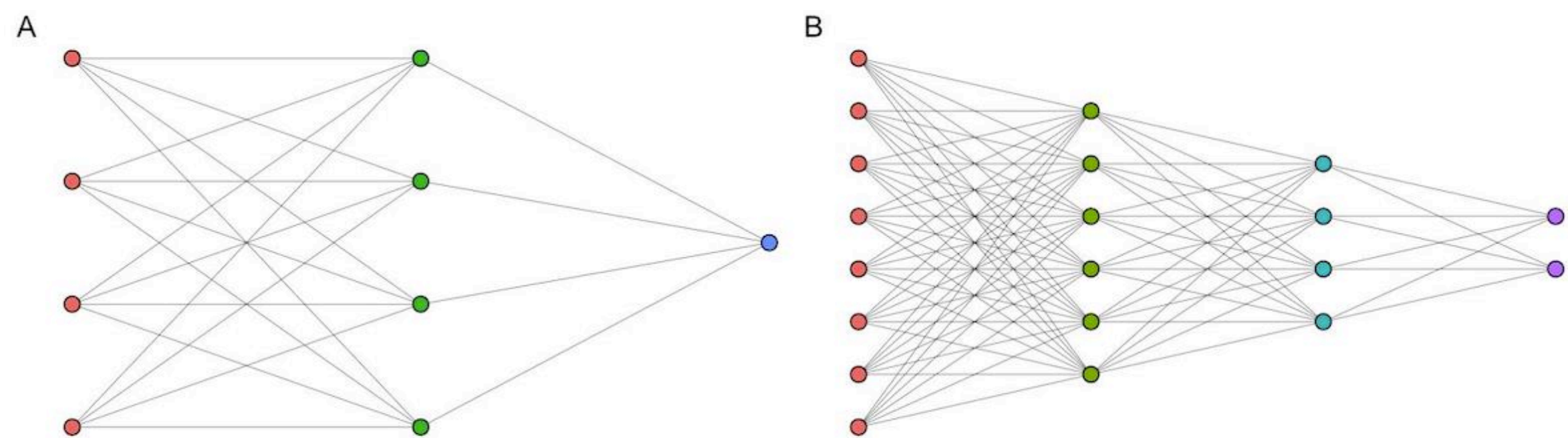
- Image classification
- Convolutional neural networks
- Hands-on: CNN for MNIST in R torch

Lecture 4: Deep learning for text

- Document classification
- Recurrent neural networks and transformers

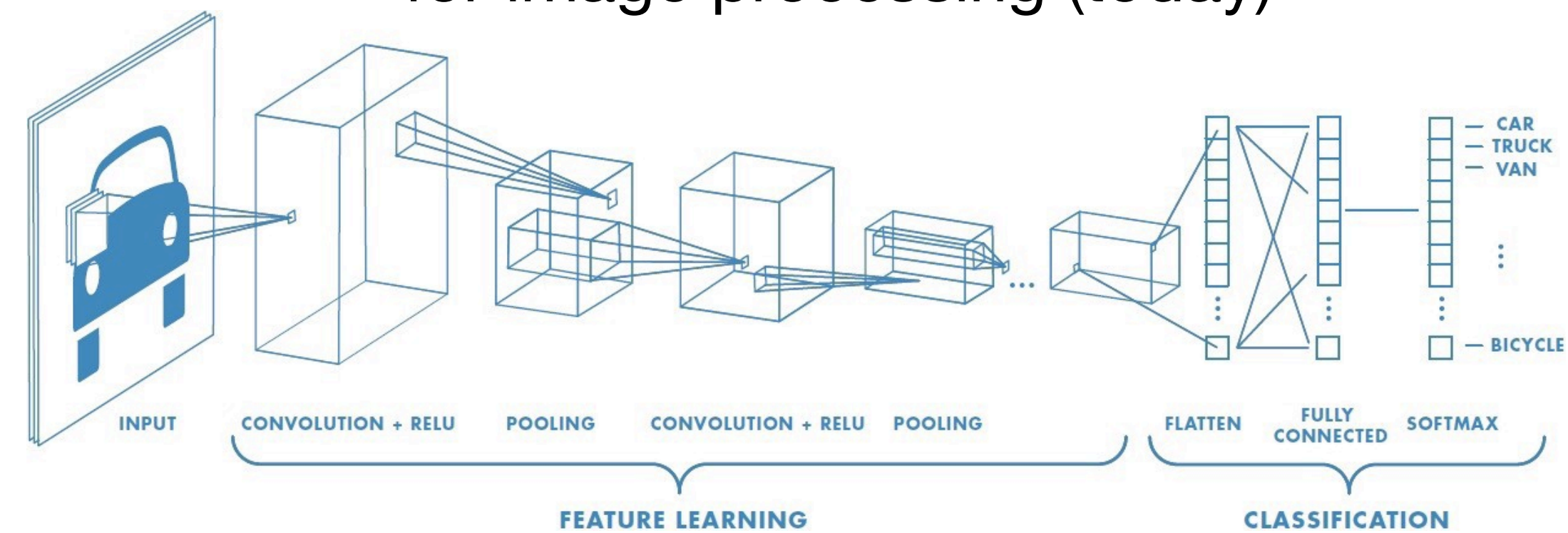
Network architectures

Fully connected architectures



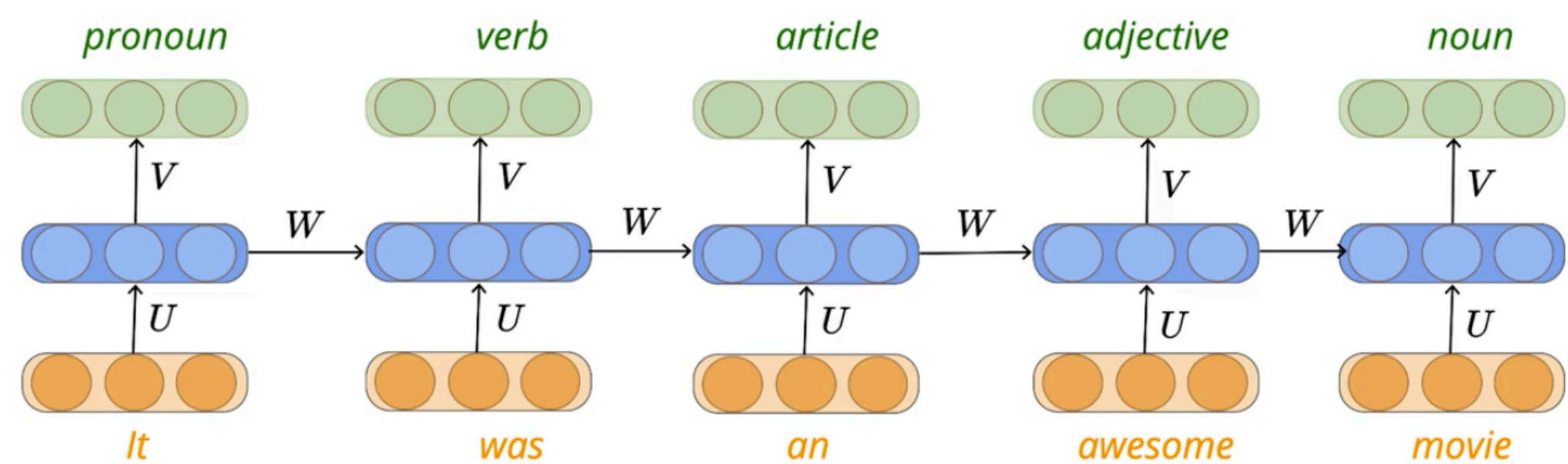
<https://community.rstudio.com/t/visualising-neural-network-architectures/41723>

Convolutional neural network (CNN) architectures for image processing (today)



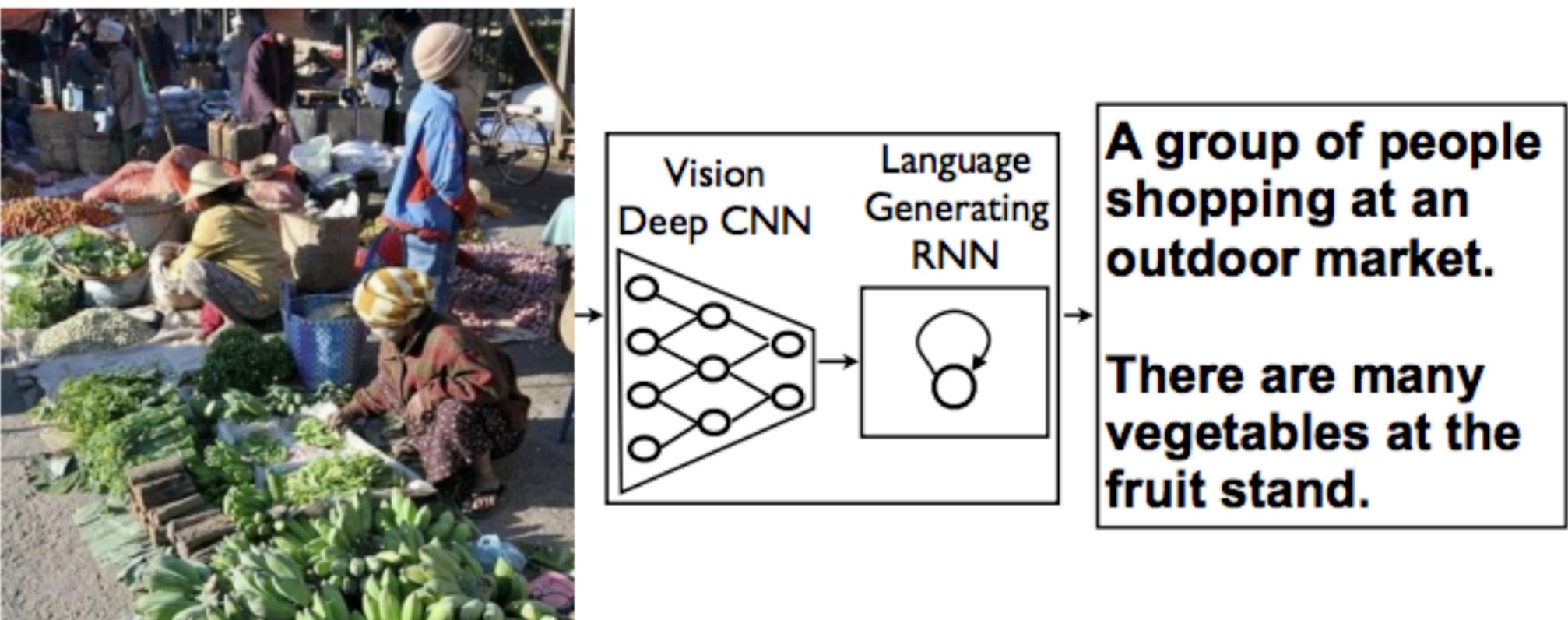
<https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53>

Recurrent neural network architectures for language processing (next lecture)



<https://towardsdatascience.com/recurrent-neural-networks-rnn-explained-the-eli5-way-3956887e8b75>

Architecture components are modular and can be composed, e.g. image captioning



https://subscription.packtpub.com/book/big_data_and_business_intelligence/9781788398060/3/ch03lvl1sec22/what-is-caption-generation

Case study: Image classification

Prototypical computer vision task:

Given an image, classify according to what object it depicts.

Challenges:

- Viewpoint variation
- Illumination
- Deformation
- Occlusion
- Background clutter
- Intraclass variation

birds



bird

cats



cat

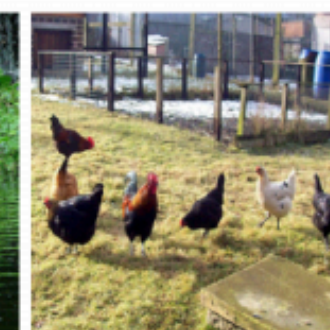
dogs



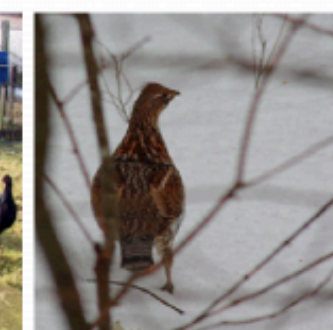
dog



flamingo



cock



ruffed grouse



quail



partridge

...



Egyptian cat



Persian cat



Siamese cat

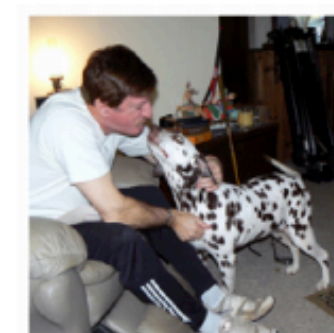


tabby



lynx

...



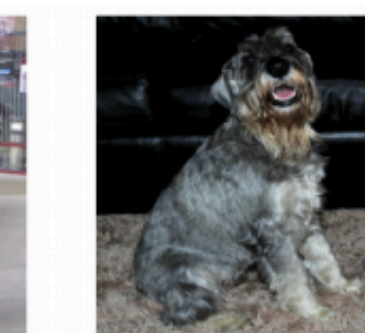
dalmatian



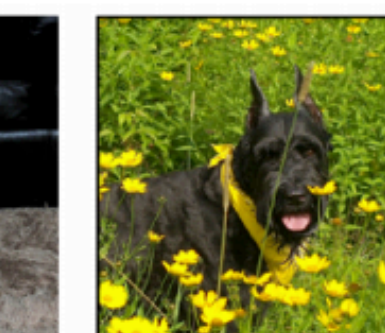
keeshond



miniature schnauzer



standard schnauzer



giant schnauzer

...

<http://ai.stanford.edu/~olga/papers/iccv13-ILSVRCanalysis.pdf>

ImageNet

A large dataset for image classification

Assembled in 2009 by downloading lots of images from the web and crowdsourcing their labels.

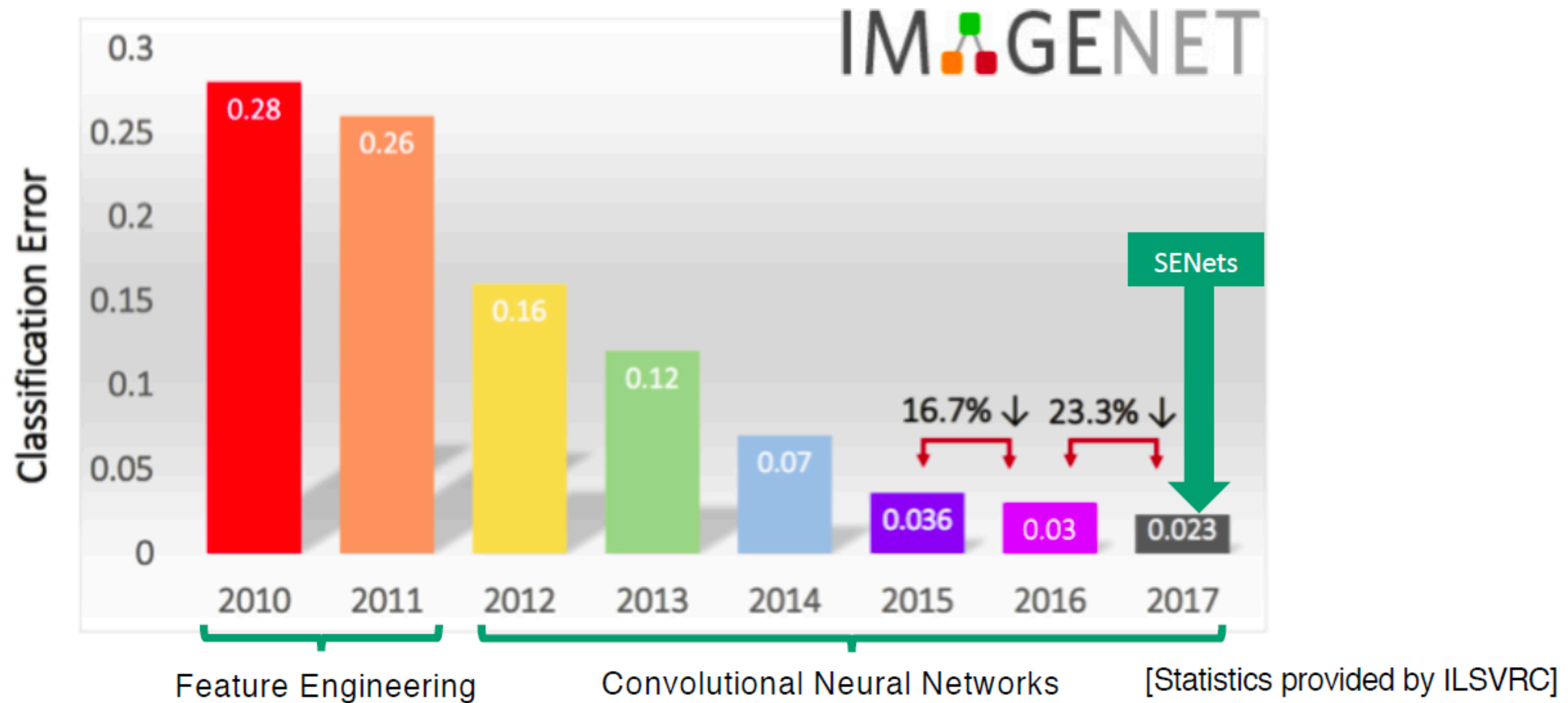
- Training set: 1.2 million images
- Test set: 100,000 images
- 1000 classes

ImageNet Large Scale Visual Recognition Challenge (ILSVRC) held annually between 2010-2017.



<https://medium.com/syncedreview/sensetime-trains-imagenet-alexnet-in-record-1-5-minutes-e944ab049b2c>

ILSVRC results over the years



Convolutional neural networks (CNNs) have dominated since 2012.

CNNs are built on image-specific properties

Figure 5.1. Images can be broken into local patterns such as edges, textures, and so on.

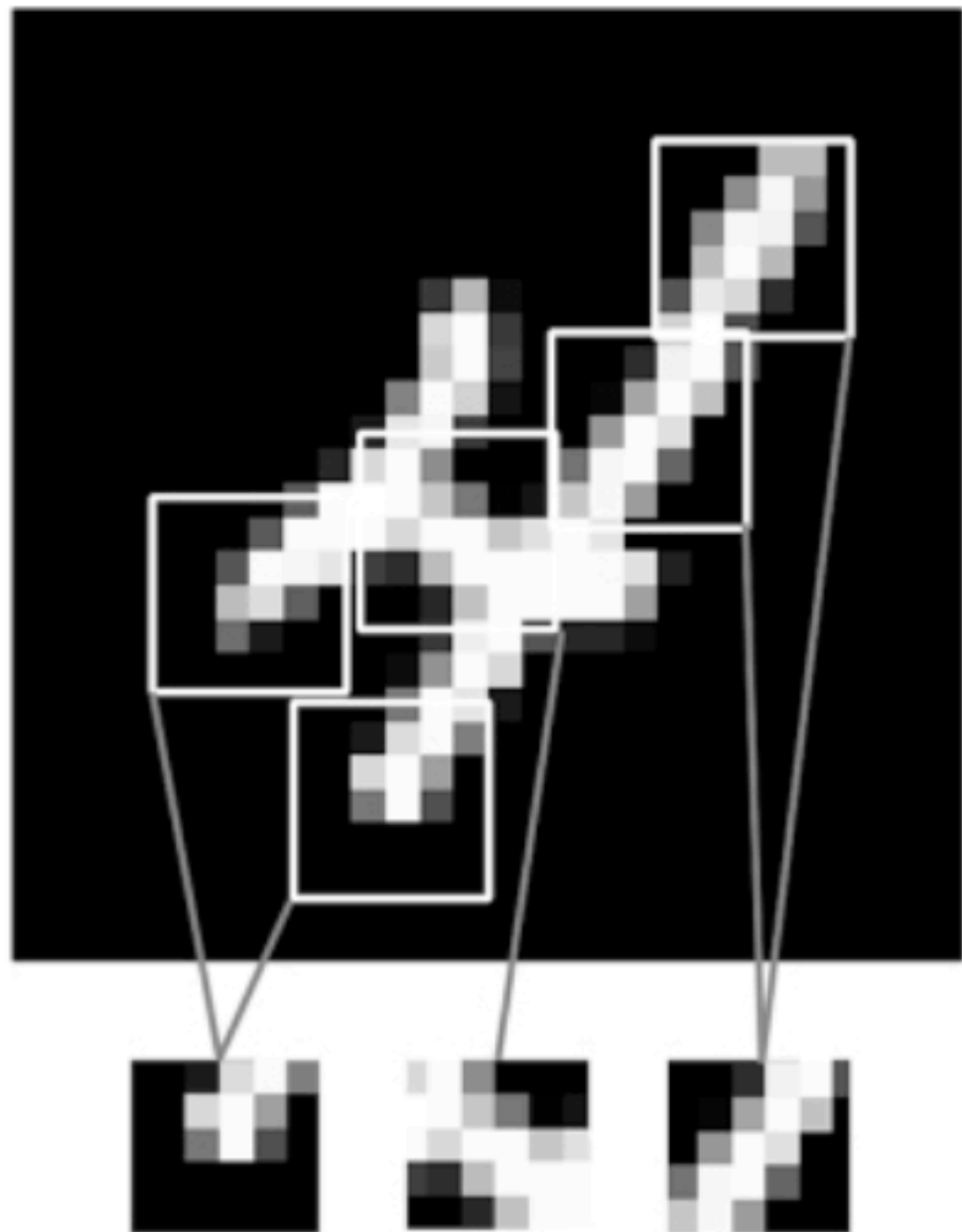
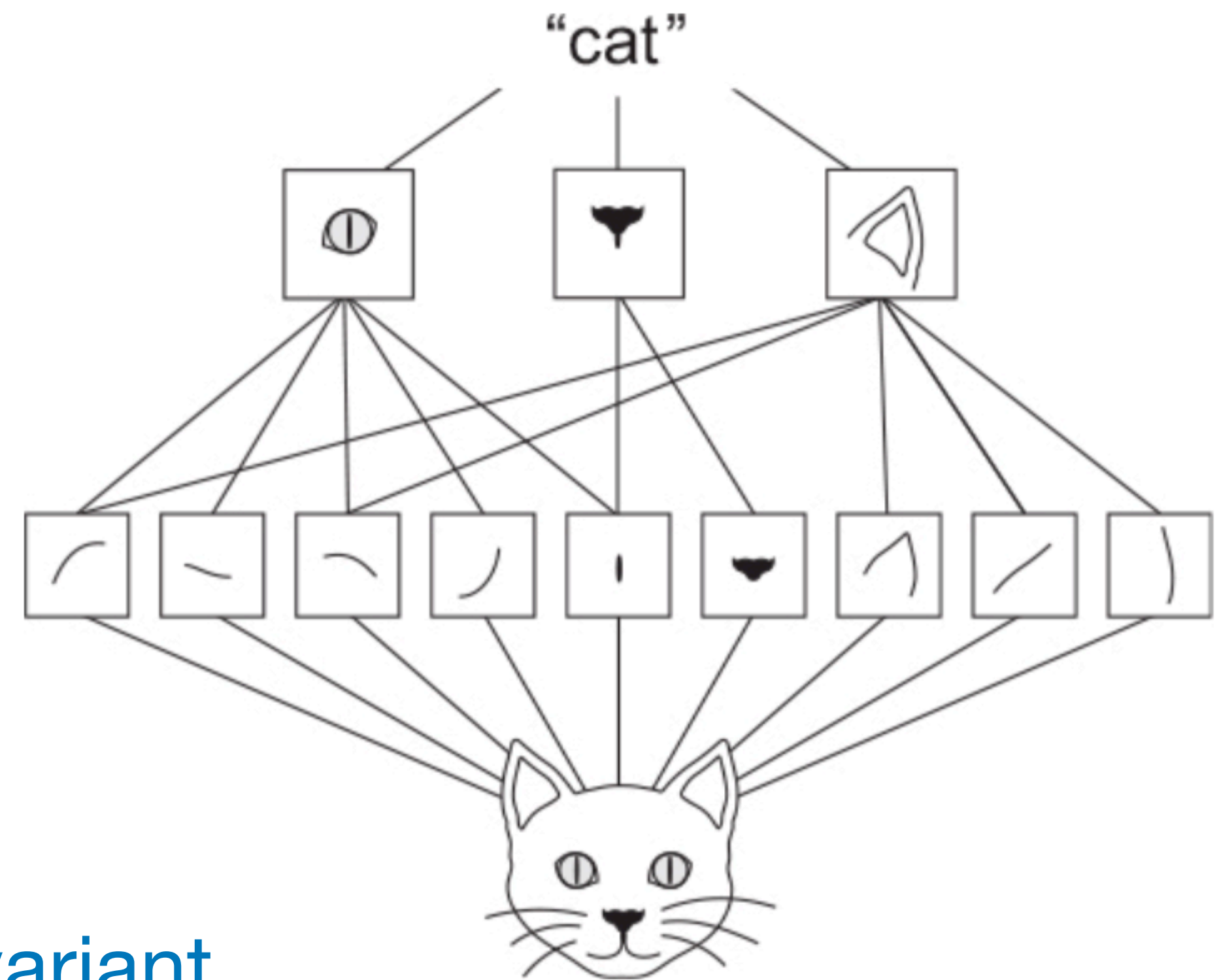


Figure 5.2. The visual world forms a spatial hierarchy of visual modules: hyperlocal edges combine into local objects such as eyes or ears, which combine into high-level concepts such as “cat.”



Patterns are

- Local
- Translation-invariant
- Hierarchical

Convolution: Searching for patterns

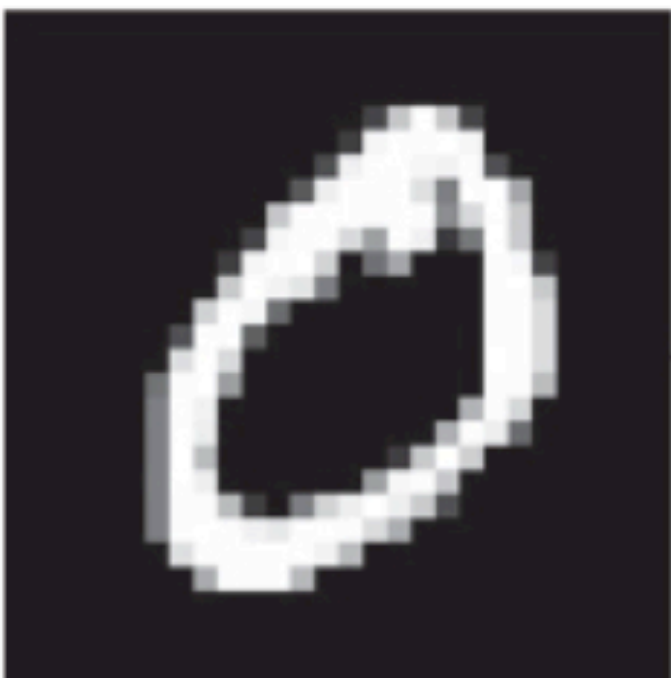
Filter (3x3)



(pattern)

0	1	2
2	2	0
0	1	2

Input image



3_0	3_1	2_2	1	0
0_2	0_2	1_0	3	1
3_0	1_1	2_2	2	3
2	0	0	2	2
2	0	0	0	1

Activation map



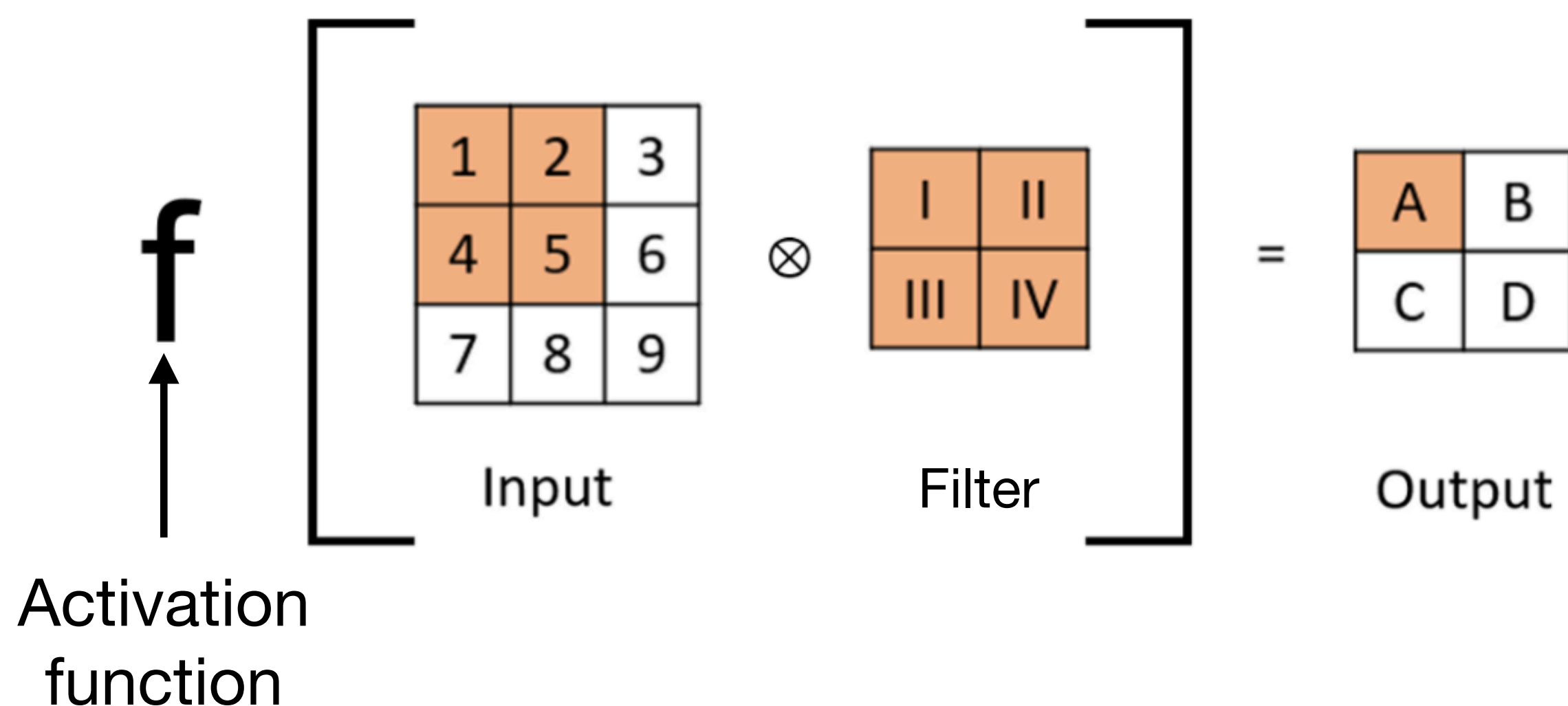
(presence of pattern)

12.0	12.0	17.0
10.0	17.0	19.0
9.0	6.0	14.0

We want to use many filters, each sensitive to a different kind of pattern.

Convolutional layer versus fully-connected layer

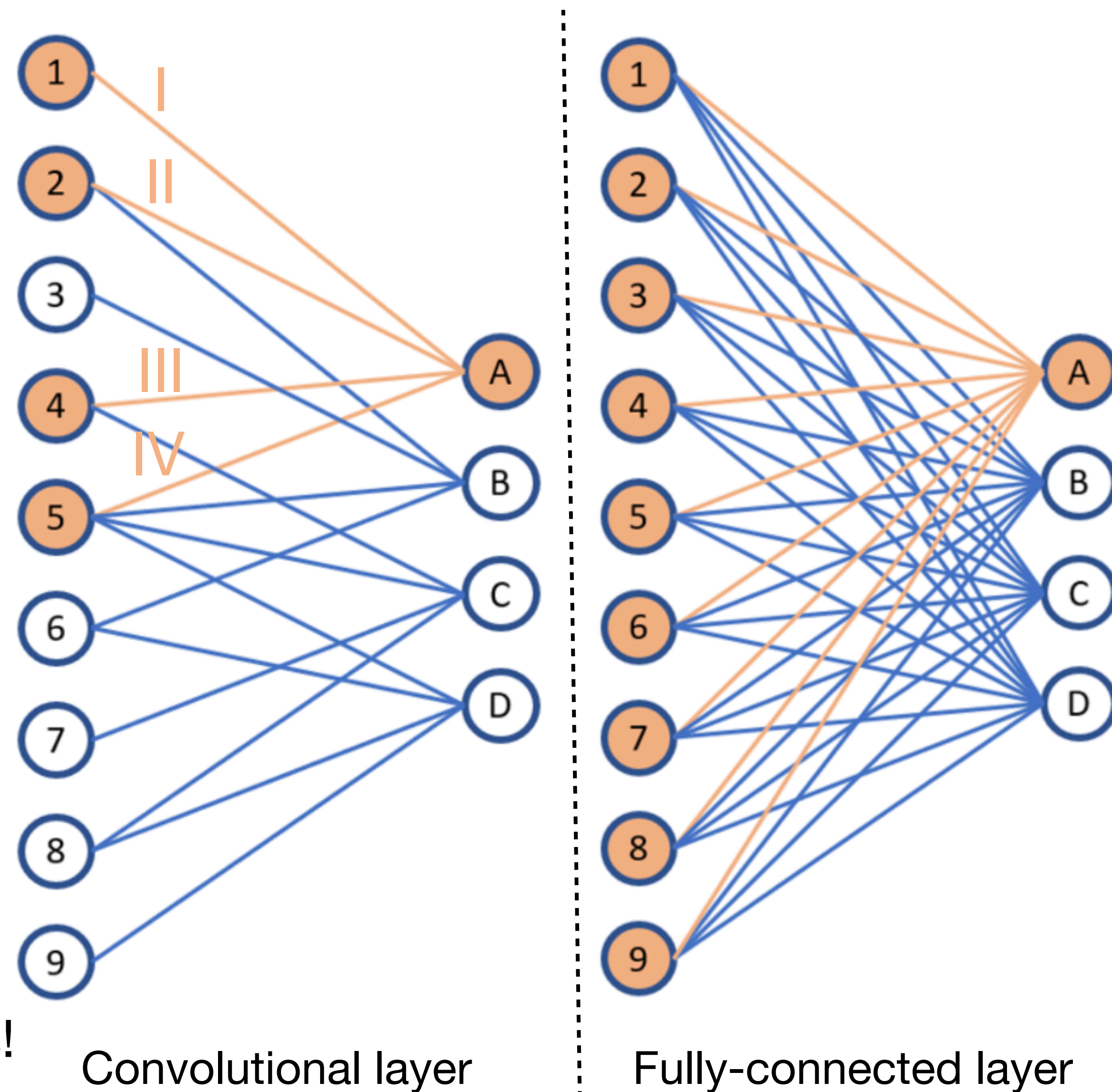
A convolutional layer can be visualized similarly to a fully-connected layer.



In a convolutional layer:

- Not all node pairs are connected with edges
- Weights (from filter) reused across edges

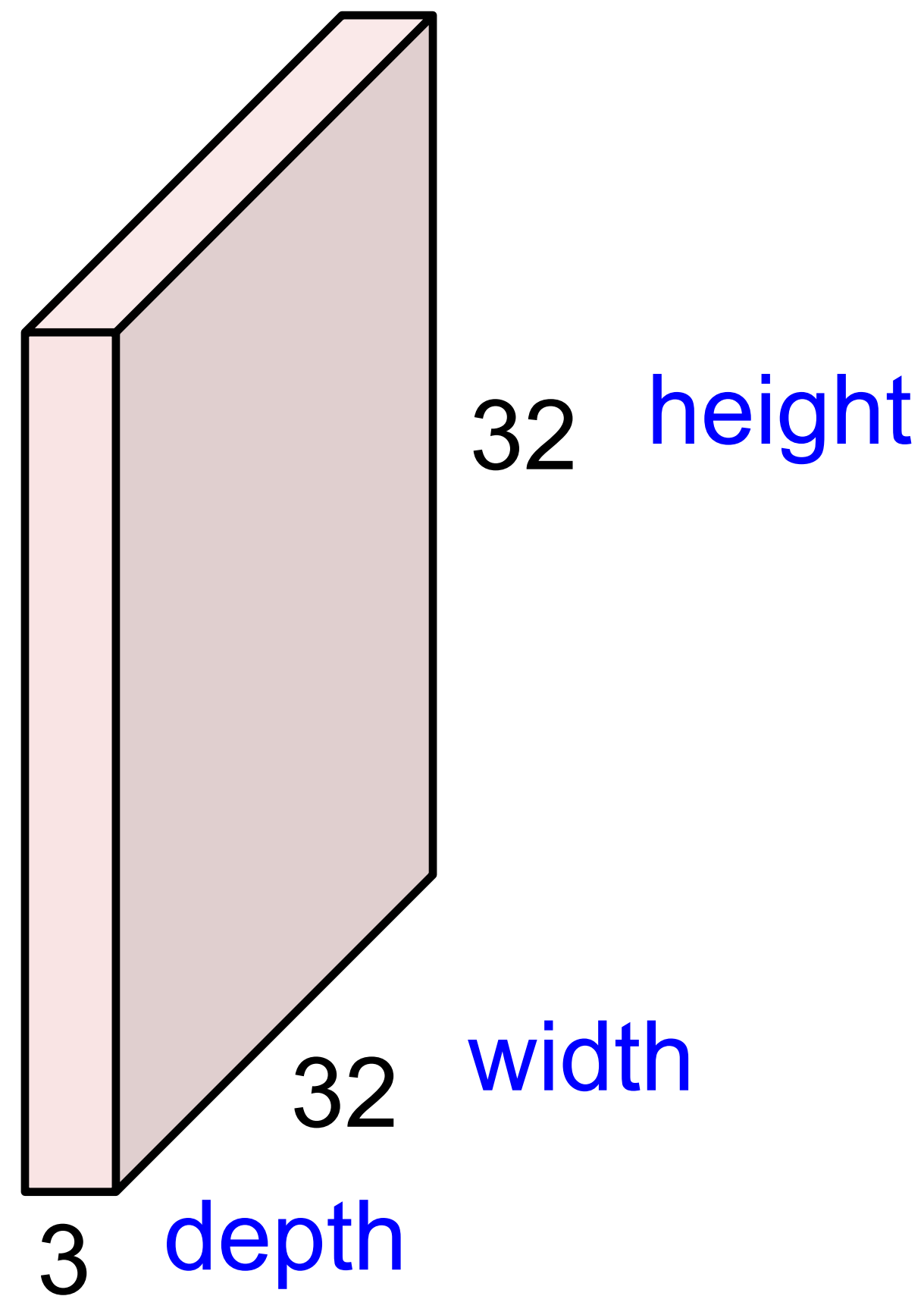
Consequence: Conv layers have fewer parameters!



Convolution Layer

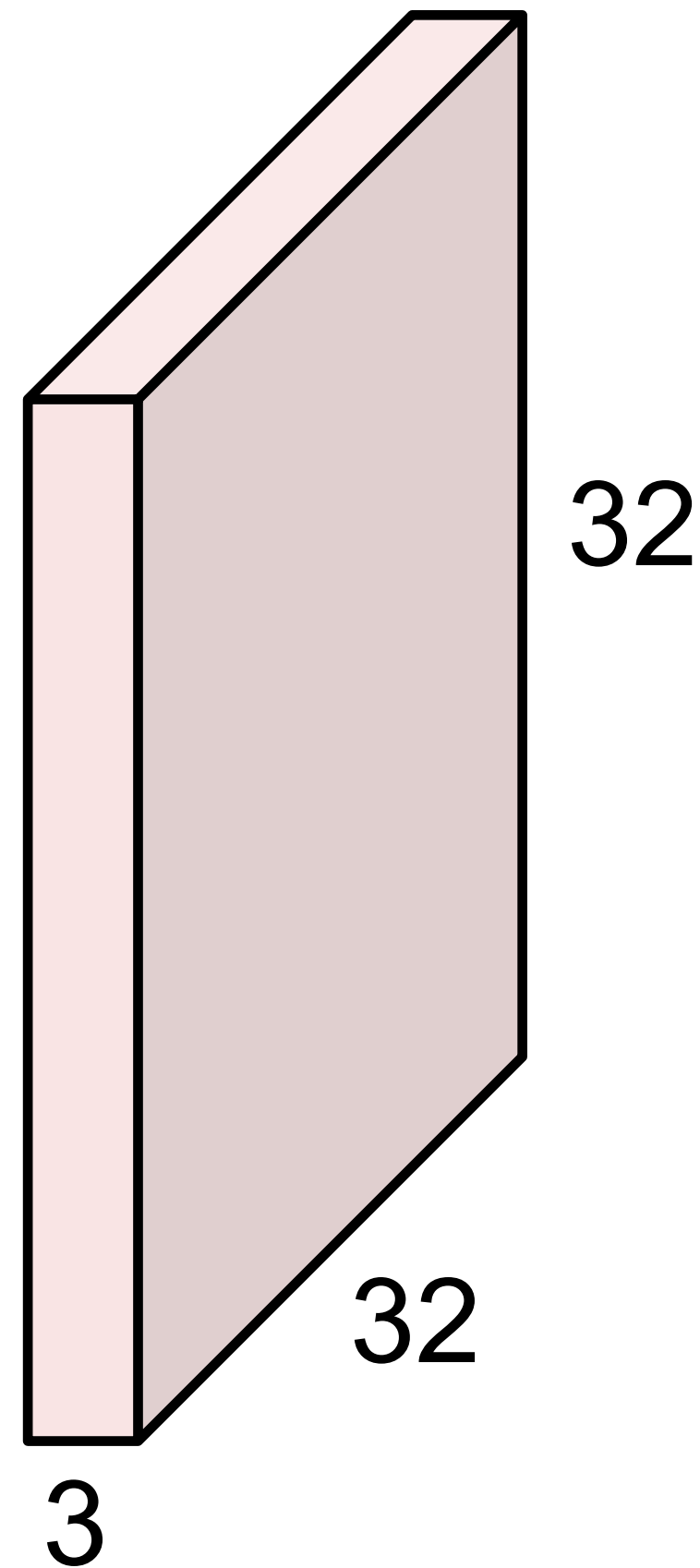
Note: This slide and several following ones are borrowed from Stanford's [CS231n](#).

32x32x3 image (images typically have red, green, and blue channels.)

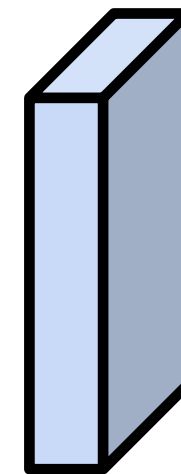


Convolution Layer

32x32x3 image



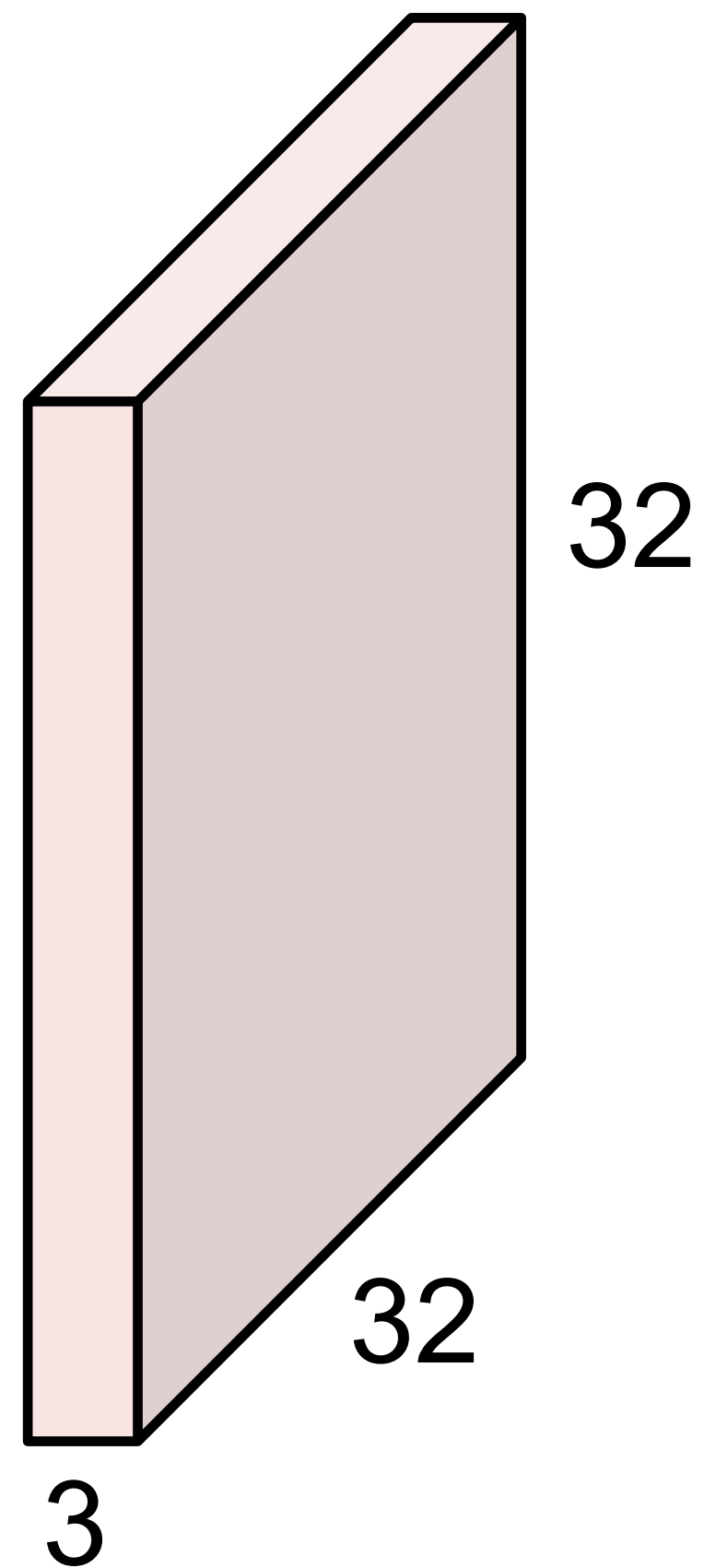
5x5x3 filter



Convolve the filter with the image
i.e. “slide over the image spatially,
computing dot products”

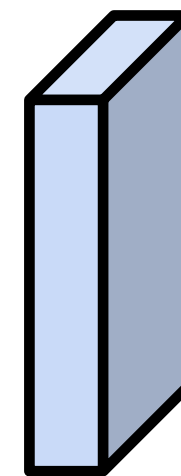
Convolution Layer

32x32x3 image



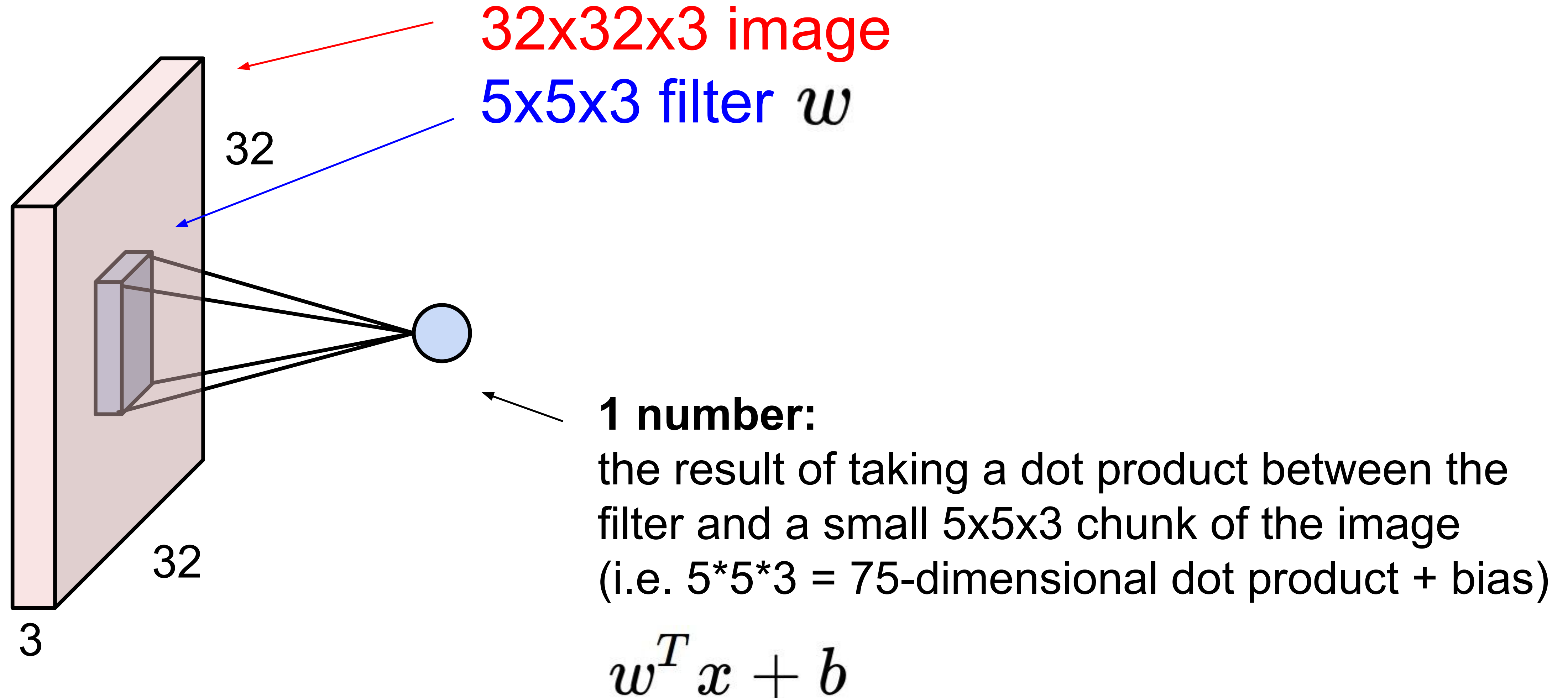
Filters always extend the full depth of the input volume

5x5x3 filter

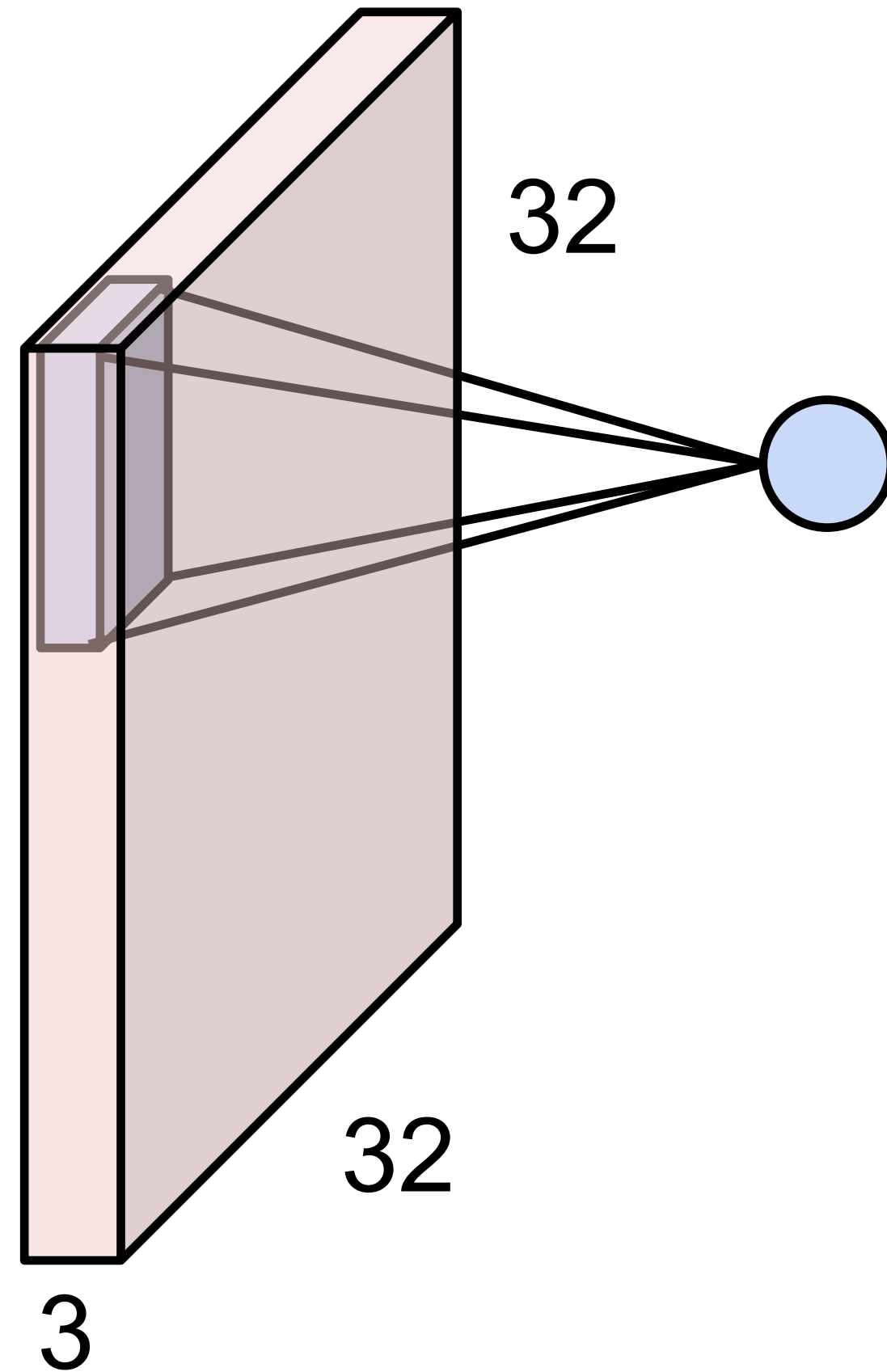


Convolve the filter with the image
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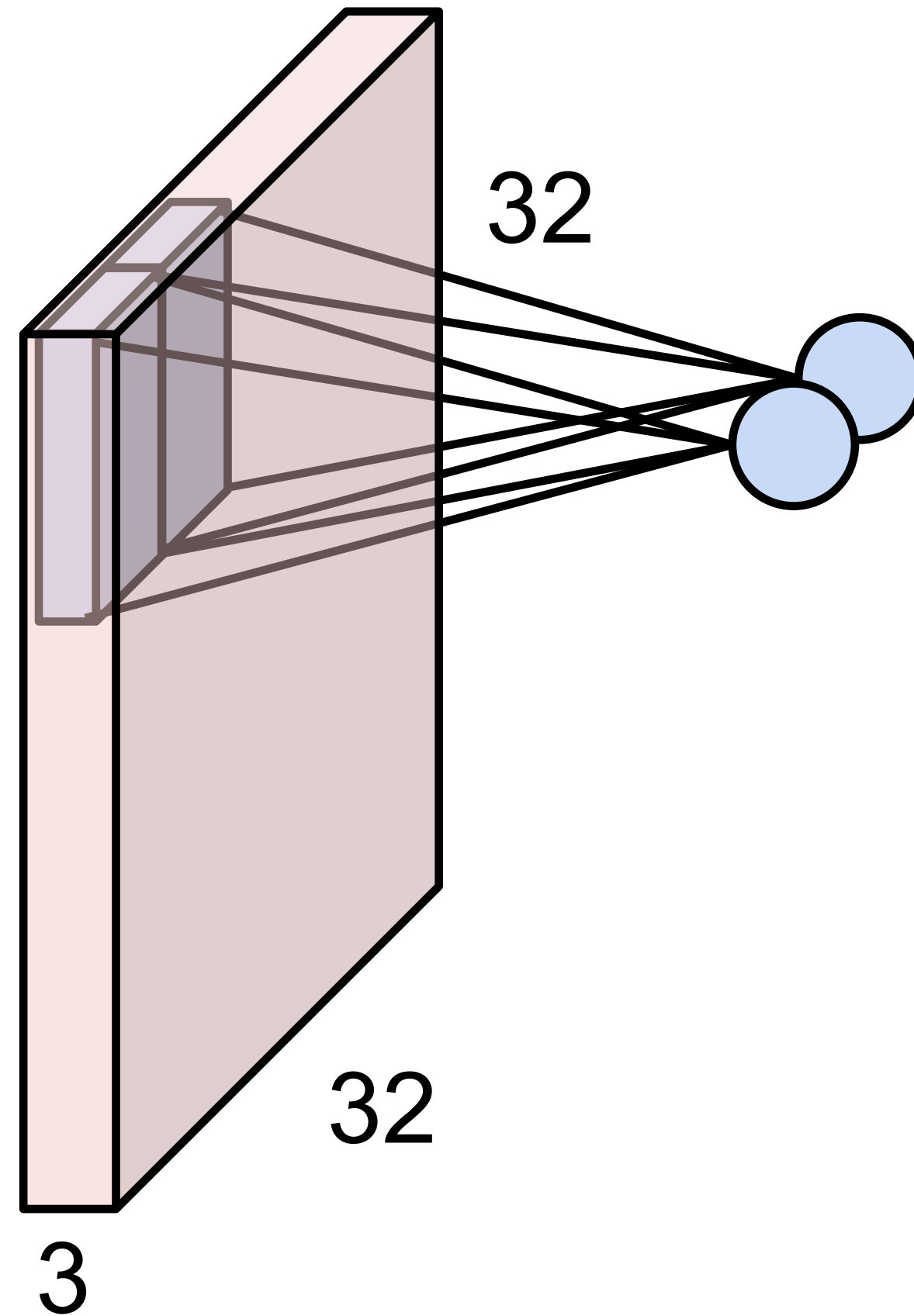
Convolution Layer



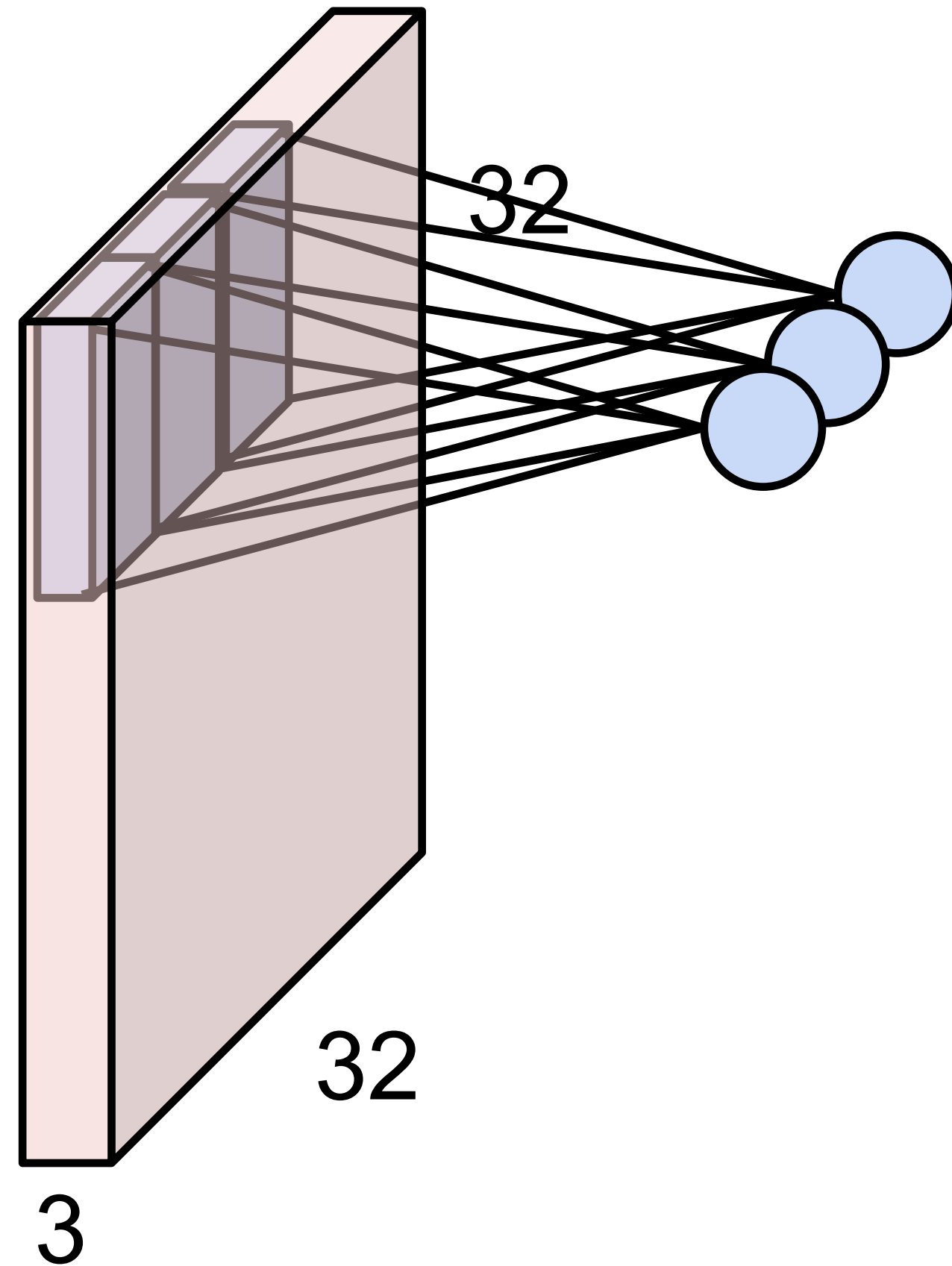
Convolution Layer



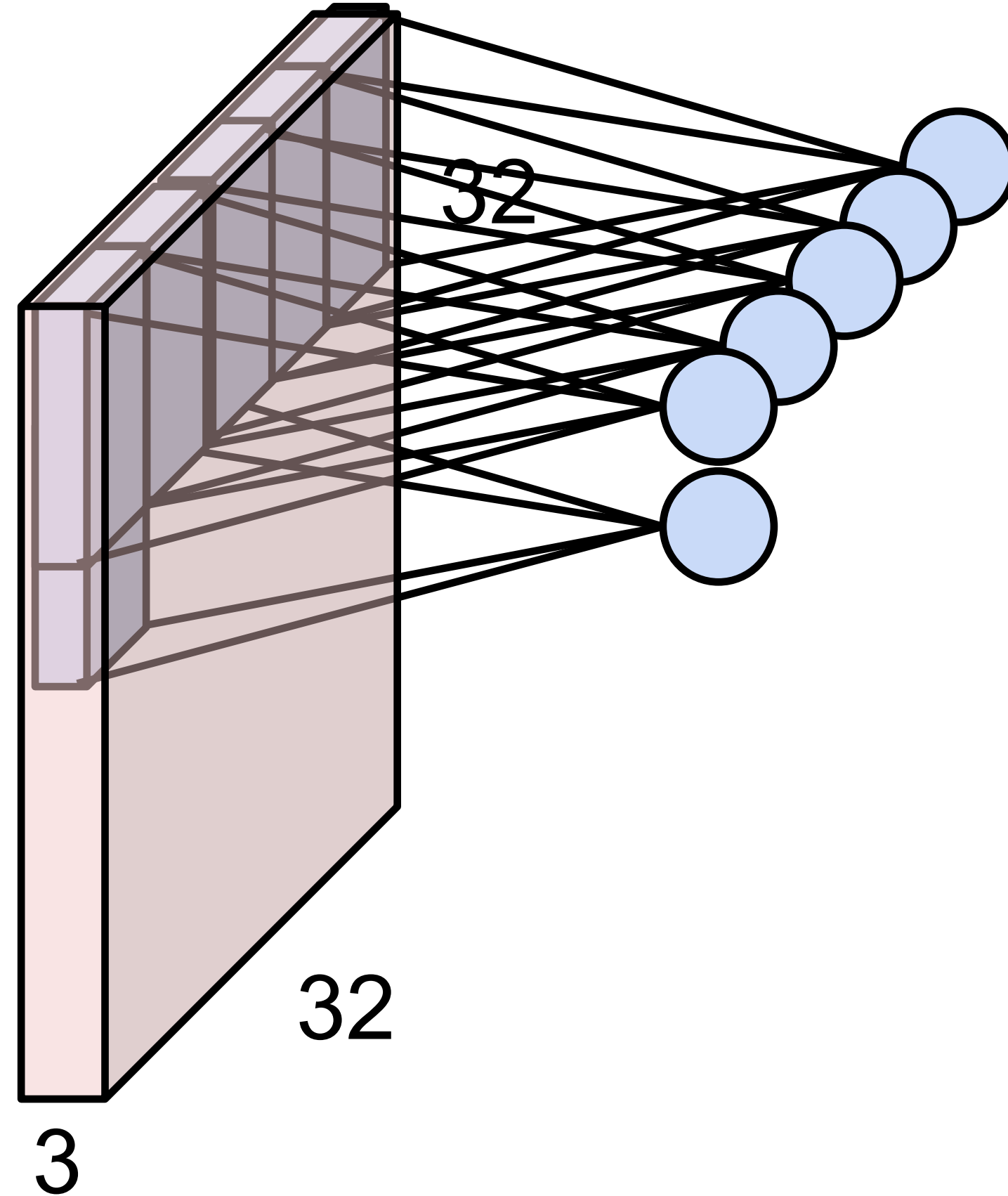
Convolution Layer



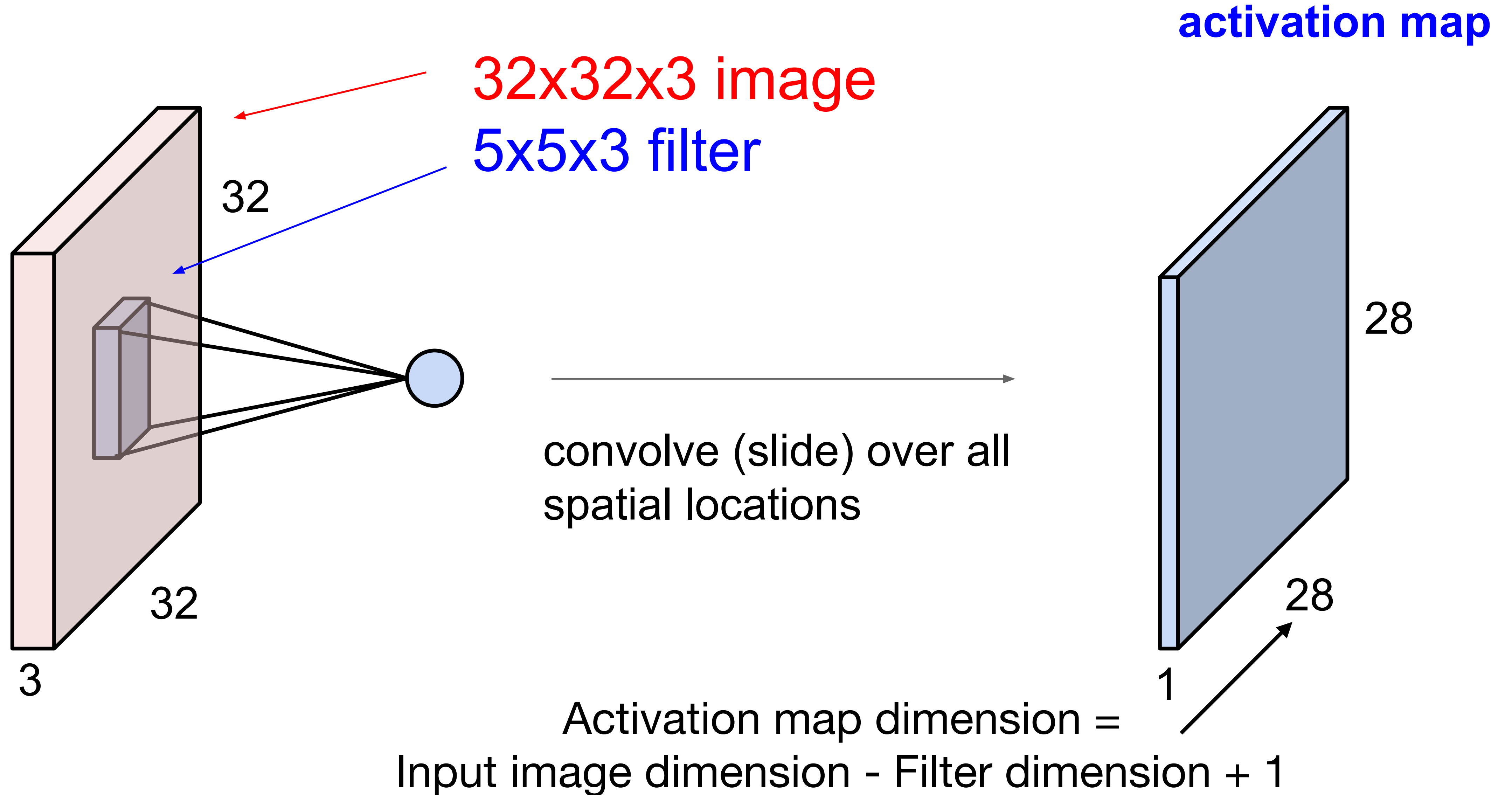
Convolution Layer



Convolution Layer



Convolution Layer

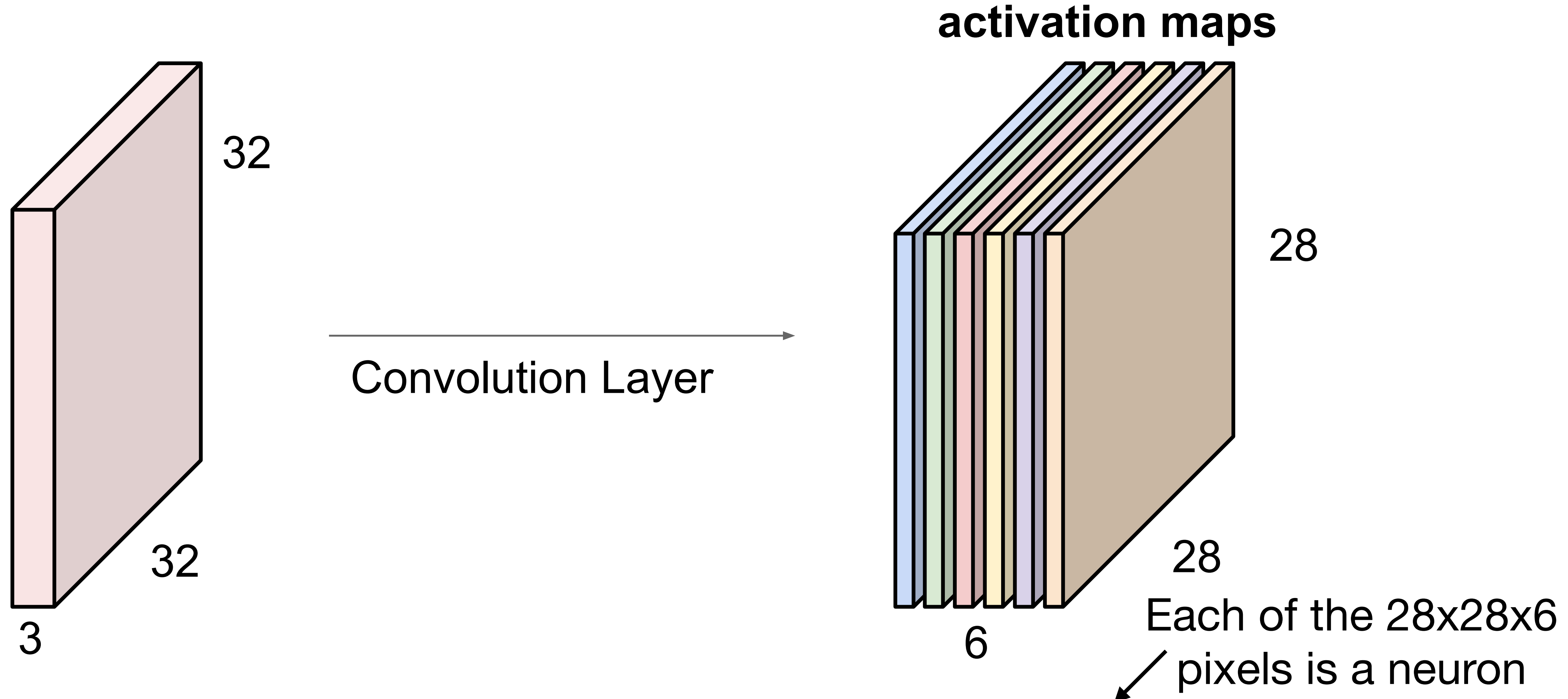


Convolution Layer

consider a second, **green** filter

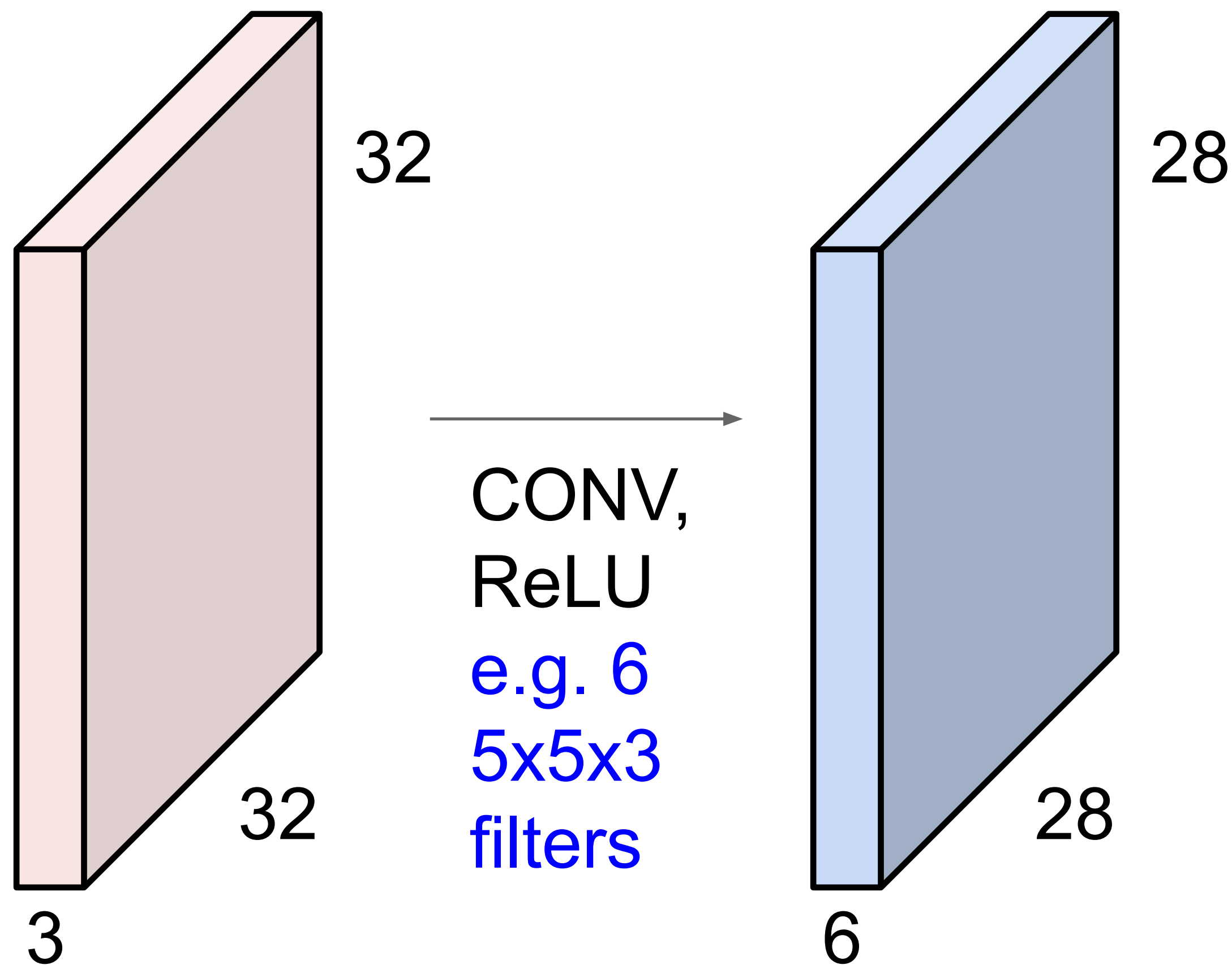


For example, if we had 6 5x5 filters, we'll get 6 separate activation maps:

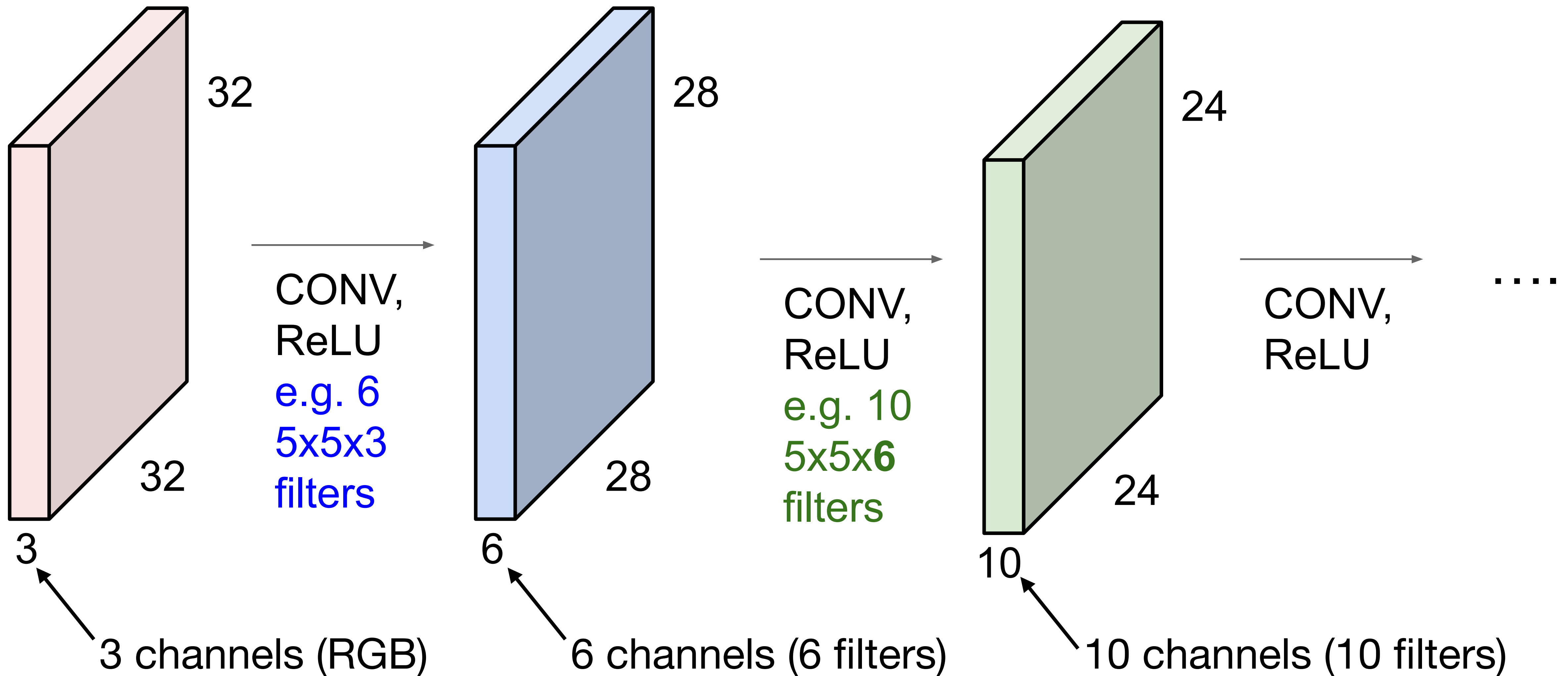


We stack these up to get a “new image” of size 28x28x6!

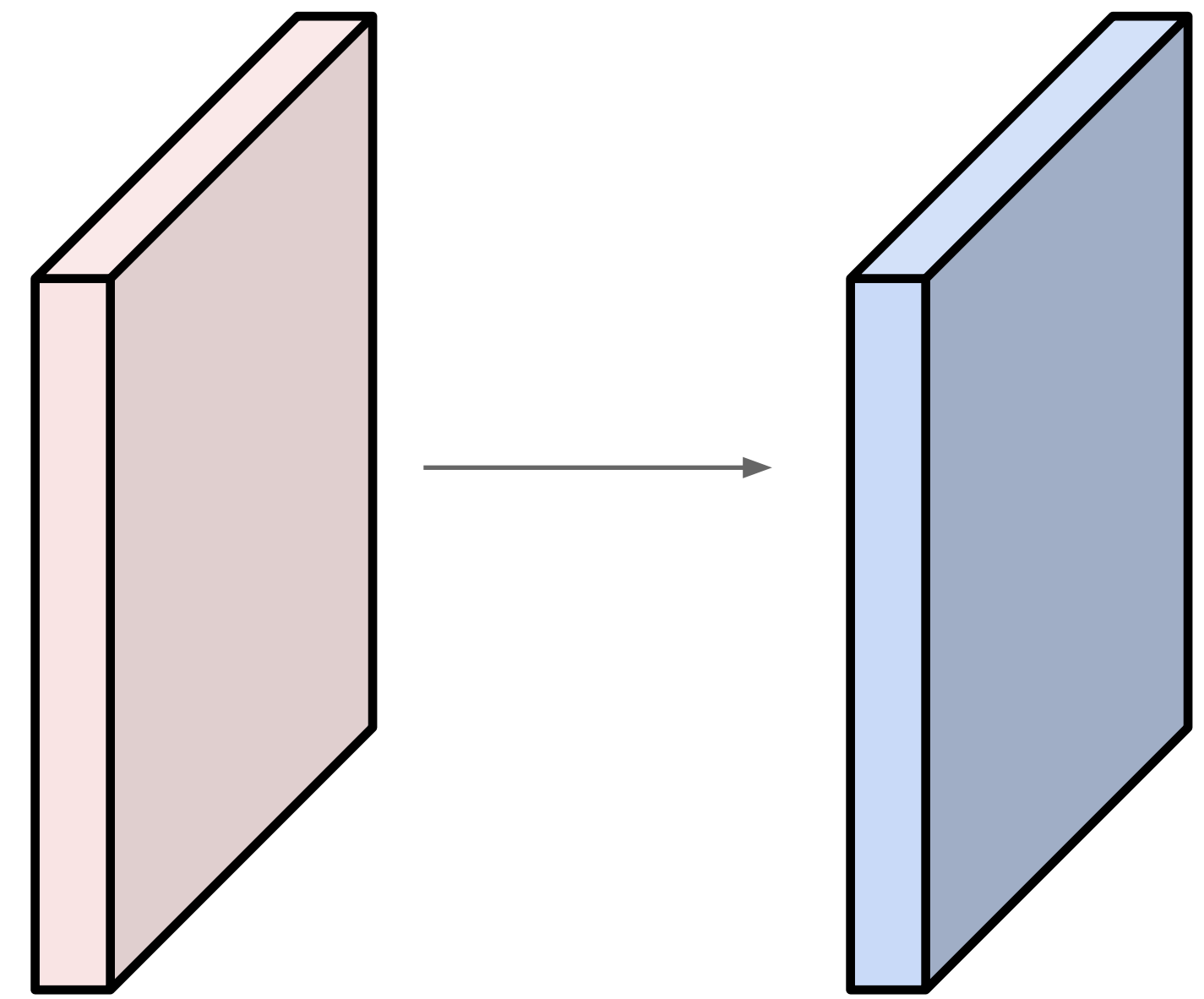
Preview: ConvNet is a sequence of Convolution Layers, interspersed with activation functions



Preview: ConvNet is a sequence of Convolution Layers, interspersed with activation functions

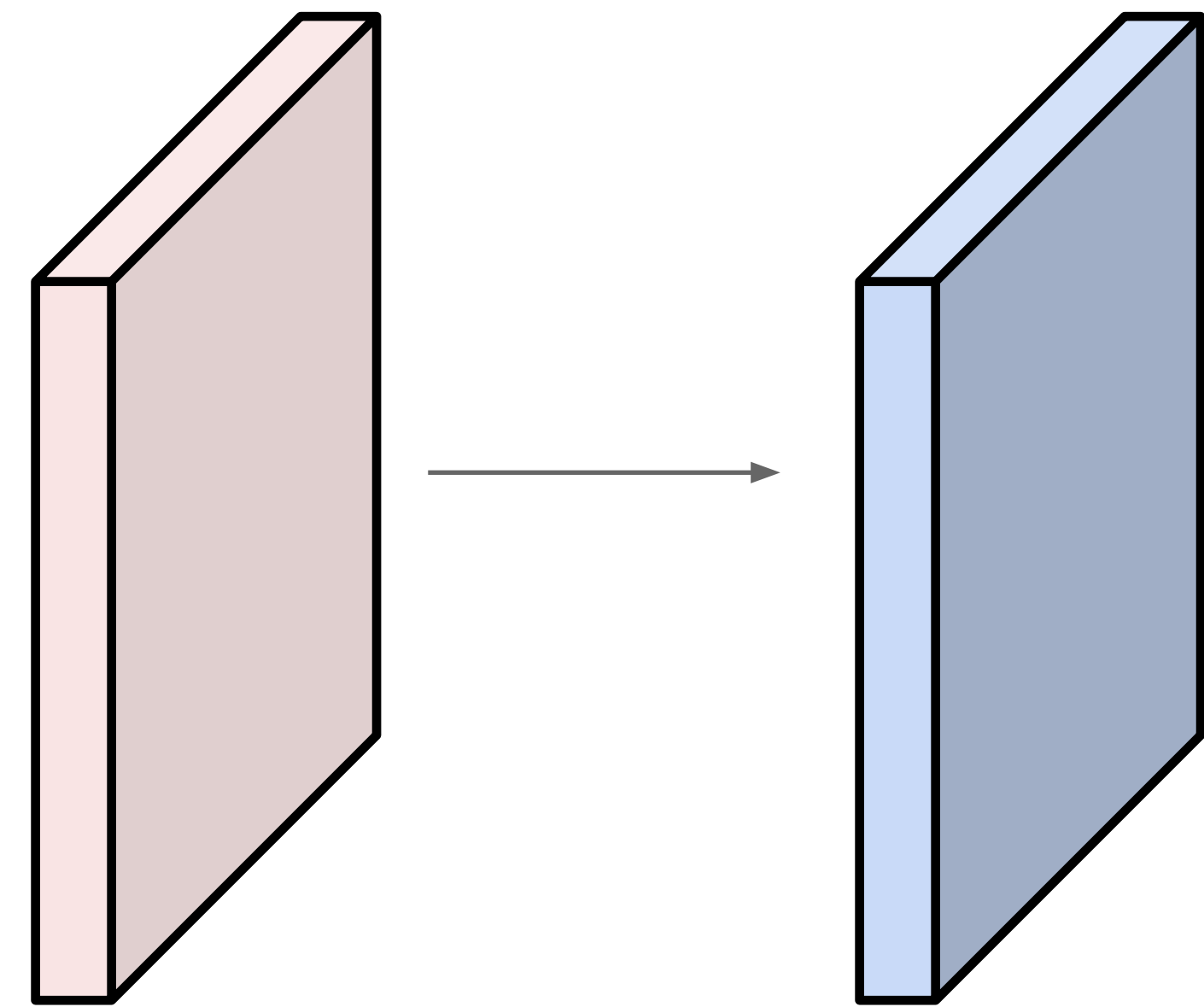


Input volume: **32x32x3**
10 5x5 filters



Number of parameters in this layer?

Input volume: **32x32x3**
10 **5x5** filters

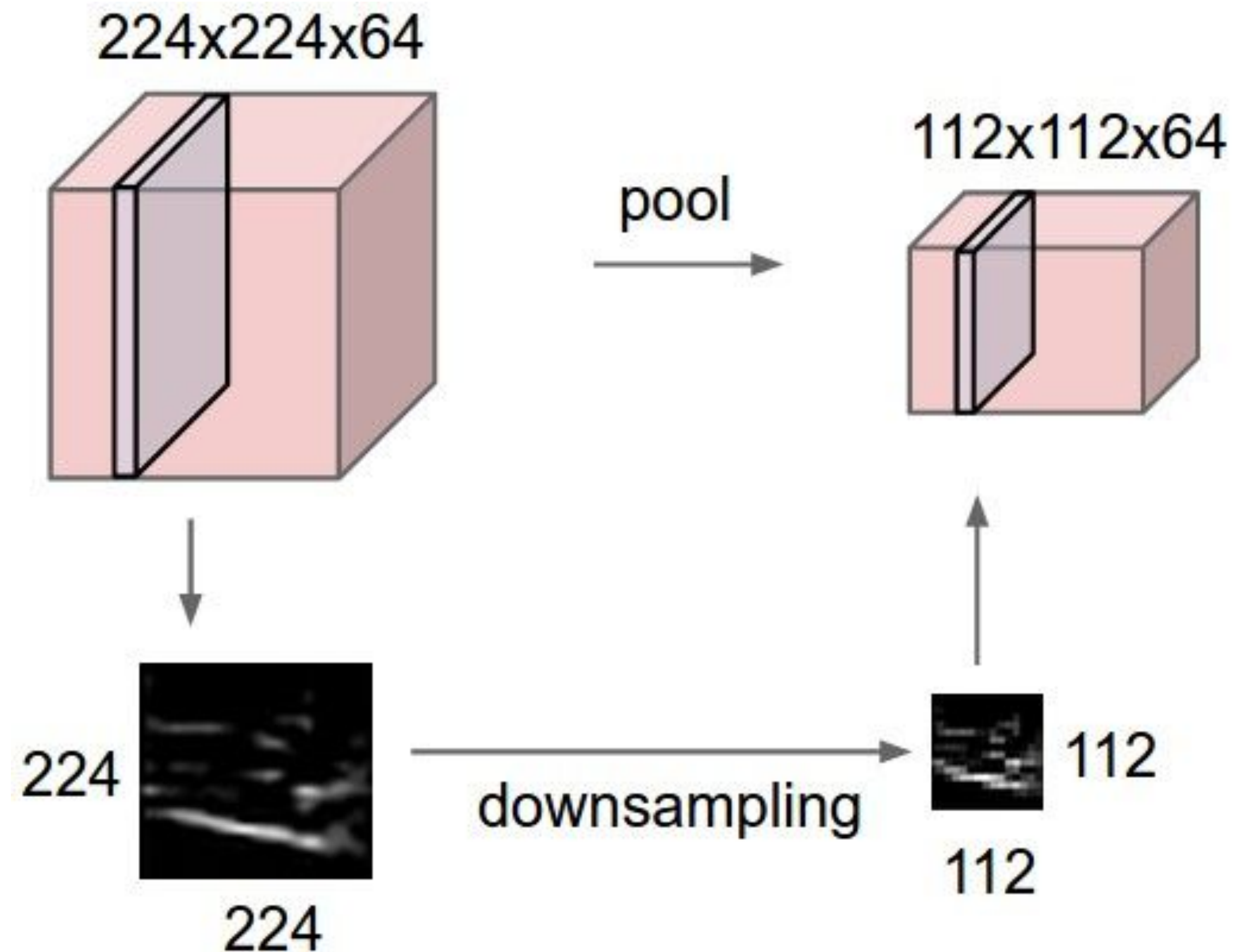


Number of parameters in this layer?
each filter has $5*5*3 + 1 = 76$ params (+1 for bias)
 $\Rightarrow 76*10 = 760$

In general, parameters in conv layer =
(filter width x filter height x input channels + 1) x number of filters.

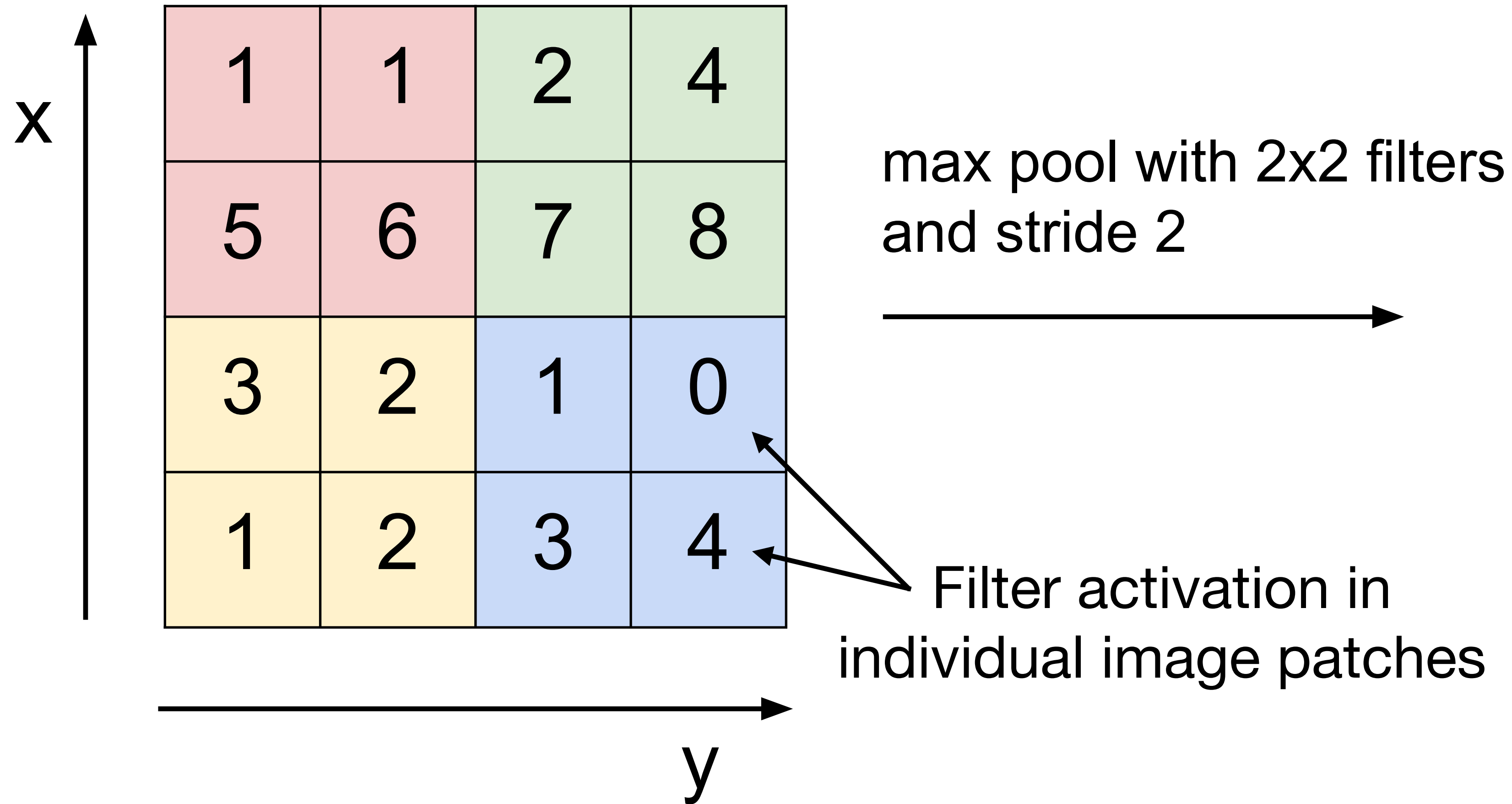
Pooling layer

- makes the representations smaller and more manageable
- operates over each activation map independently:



MAX POOLING

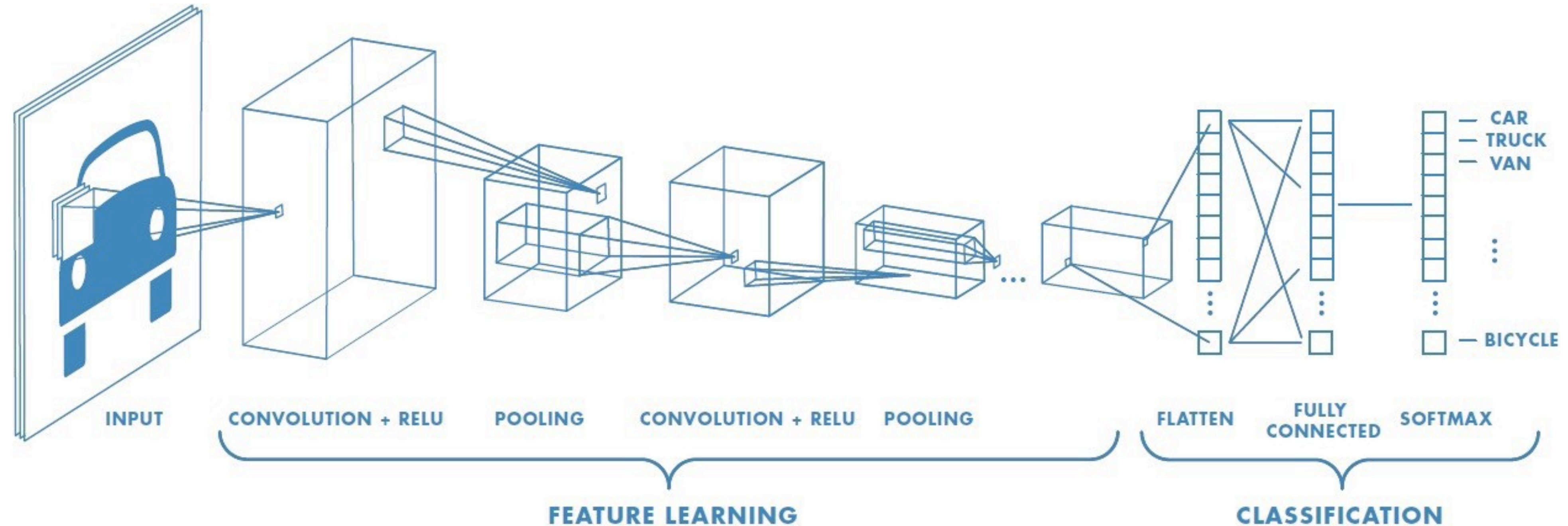
Single depth slice



6	8
3	4

Maximum filter activation across adjacent patches

Convolutional neural networks



<https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53>

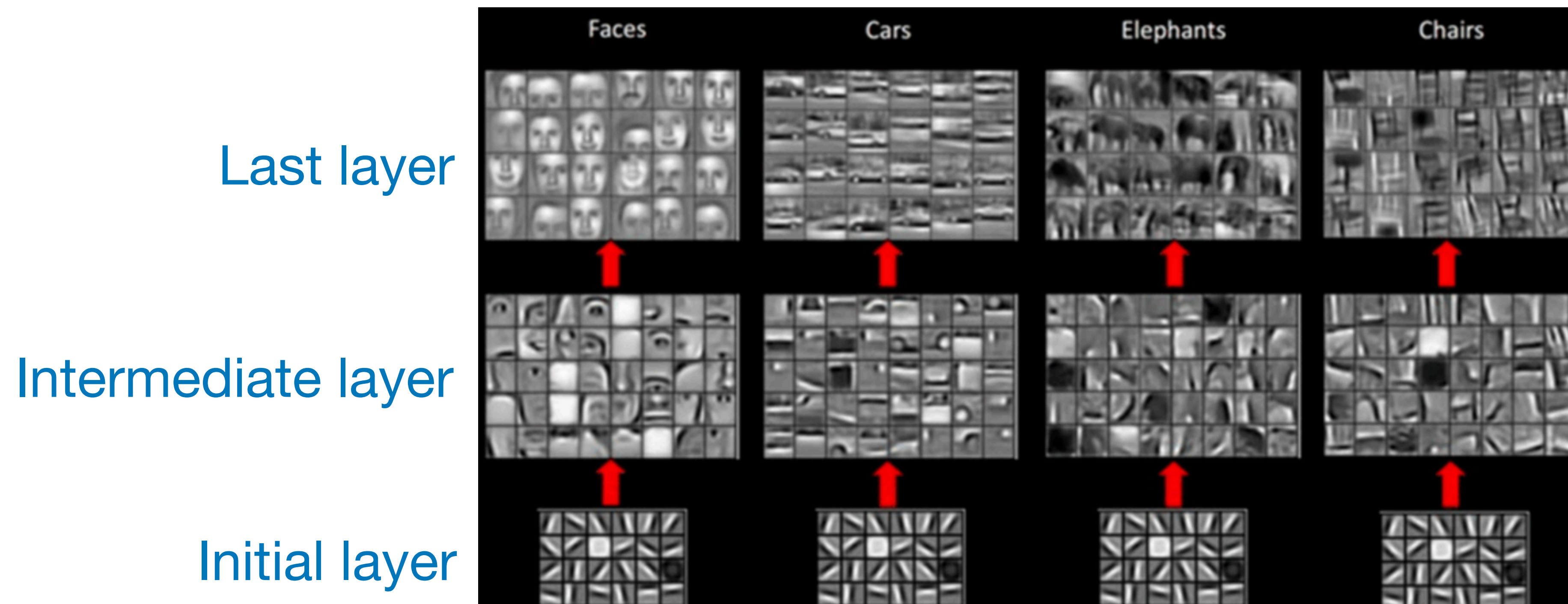
A CNN stacks together several alternating convolution and pooling layers, followed by a fully connected layer and a softmax output.

Filters, weights in fully connected layer, and biases learned by optimizing cross-entropy loss via stochastic gradient descent.

Interpreting the filters learned by a CNN

Use neural network for binary classification, e.g. faces versus not faces, cars versus not cars, etc.

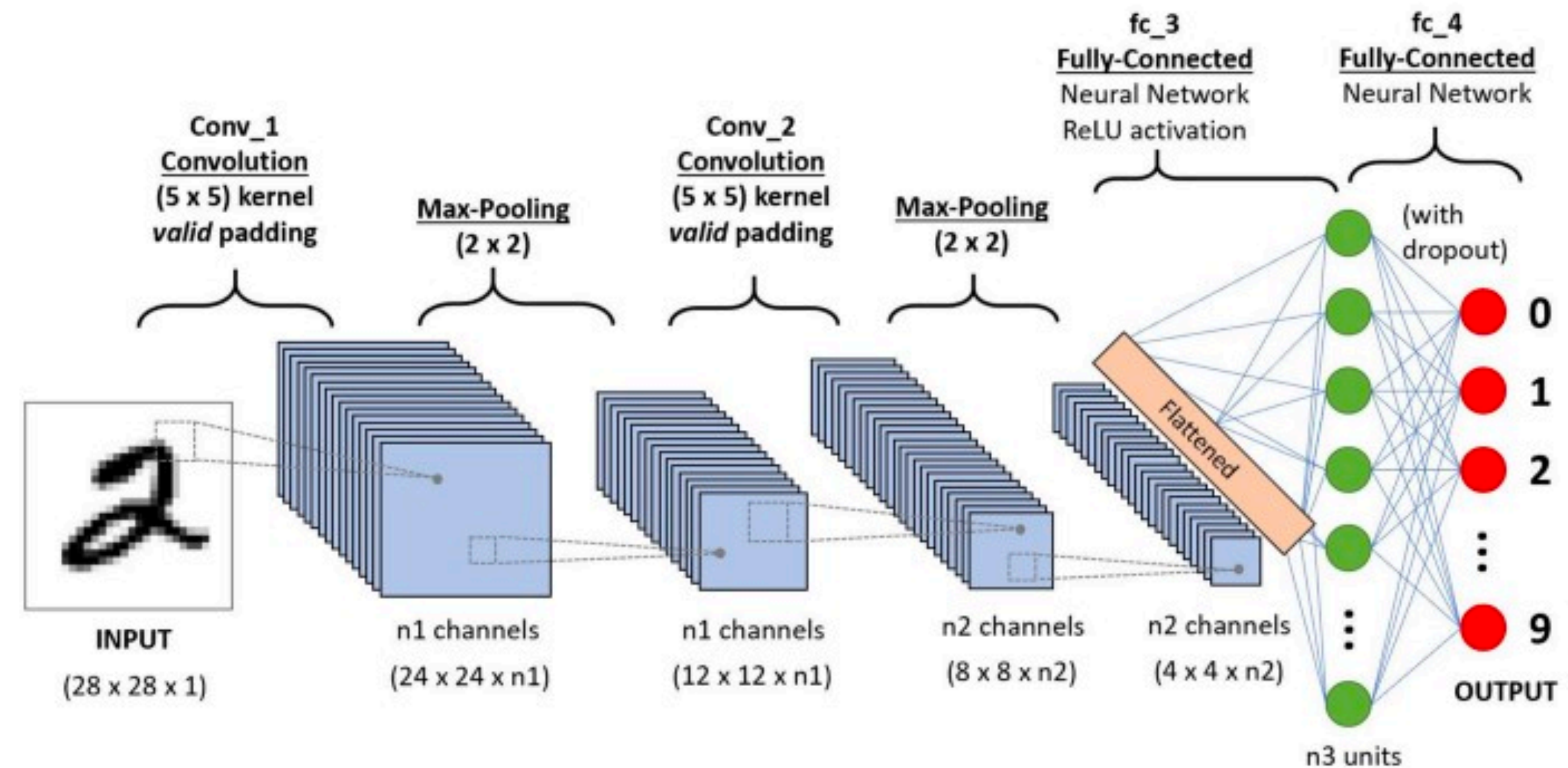
For each neuron at each layer, find input image that activates it most strongly.



The original CNN architecture

LeNet architecture for hand-written digit recognition (1989).

Idea existed decades ago but data and computing power only became available in 2010s.



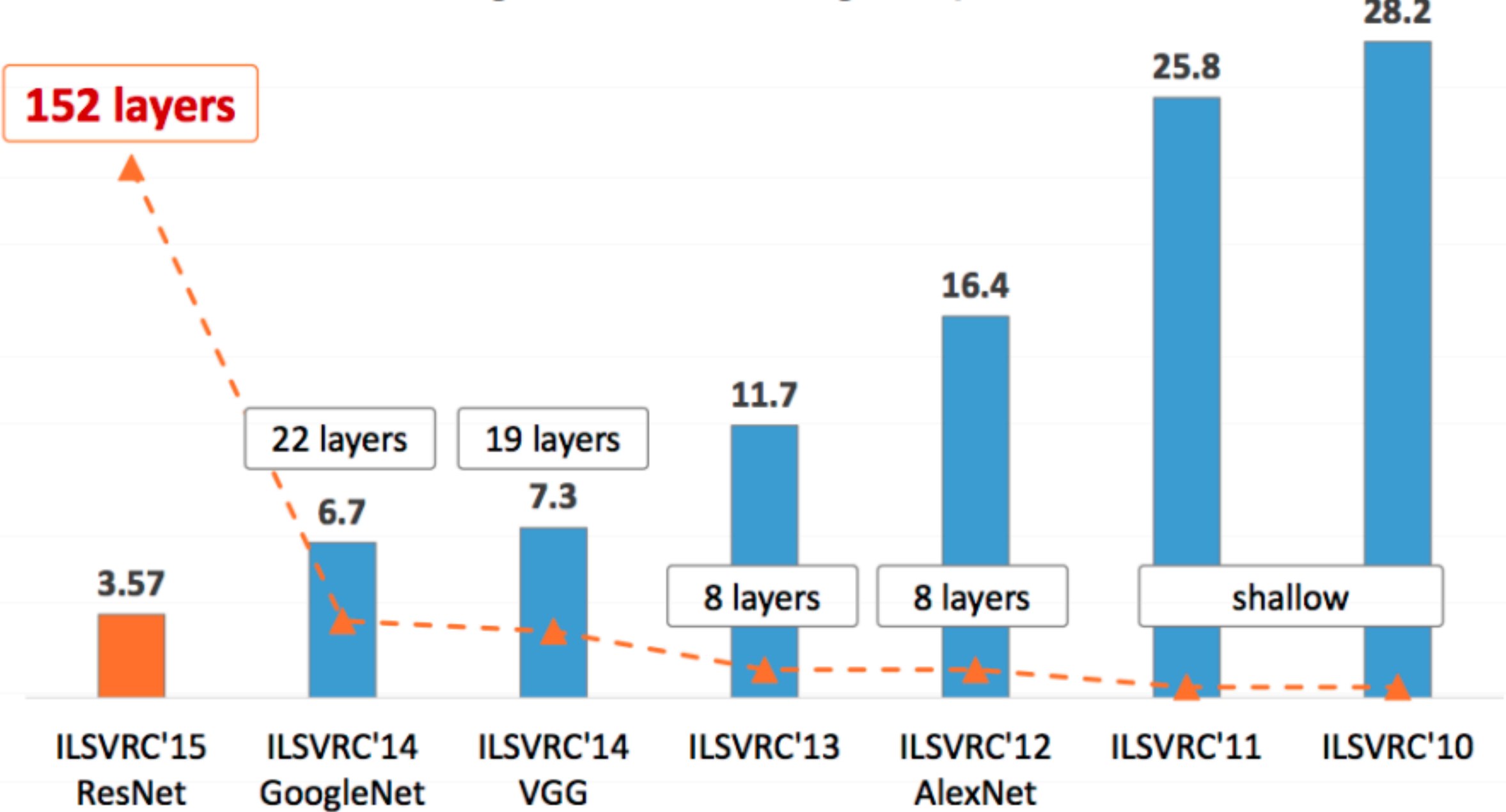
<https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53>



Yann LeCun

Modern CNN architectures

Classification: ImageNet Challenge top-5 error



<https://medium.com/@RaghavPrabhu/cnn-architectures-lexnet-alexnet-vgg-googlenet-and-resnet-7c81c017b848>

Model	Size (M)	Top-1/top-5 error (%)	# layers	Model description
AlexNet	238	41.00/18.00	8	5 conv + 3 fc layers
VGG-16	540	28.07/9.33	16	13 conv + 3 fc layers
VGG-19	560	27.30/9.00	19	16 conv + 3 fc layers
GoogLeNet	40	29.81/10.04	22	21 conv + 1 fc layers
ResNet-50	100	22.85/6.71	50	49 conv + 1 fc layers
ResNet-152	235	21.43/3.57	152	151 conv + 1 fc layers

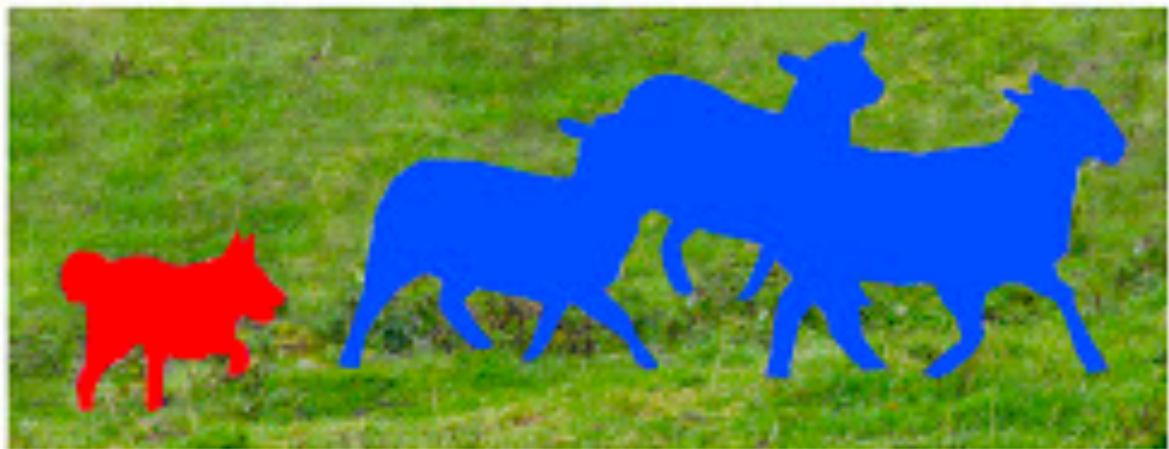
https://www.researchgate.net/figure/The-comparison-of-different-CNN-architectures-on-model-size-classification-error-rate_tbl1_320199404

CNNs are getting progressively deeper with time.

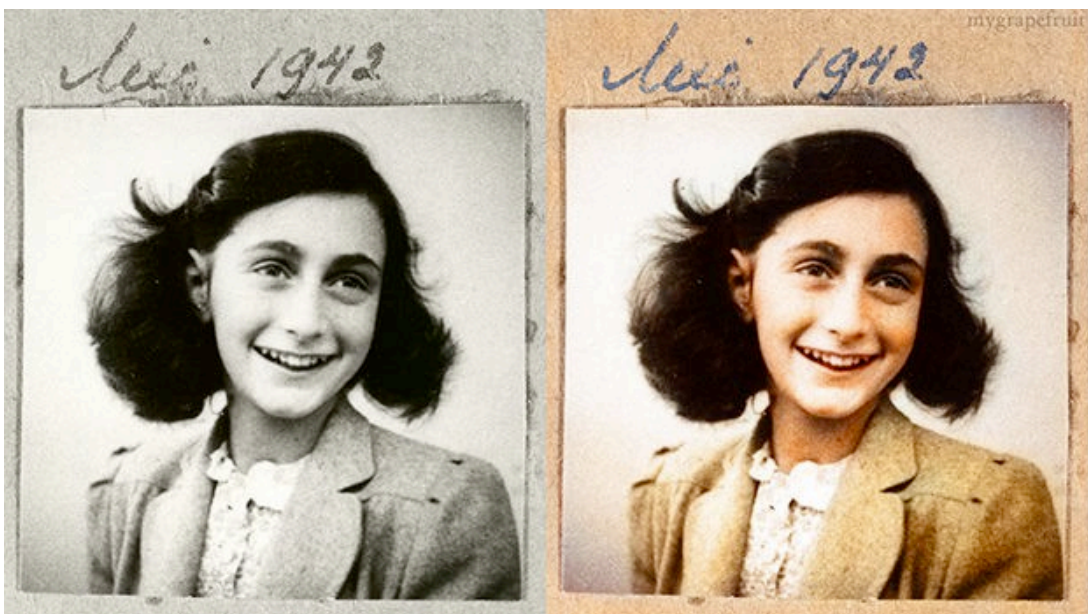
Other applications of CNNs



Image Recognition

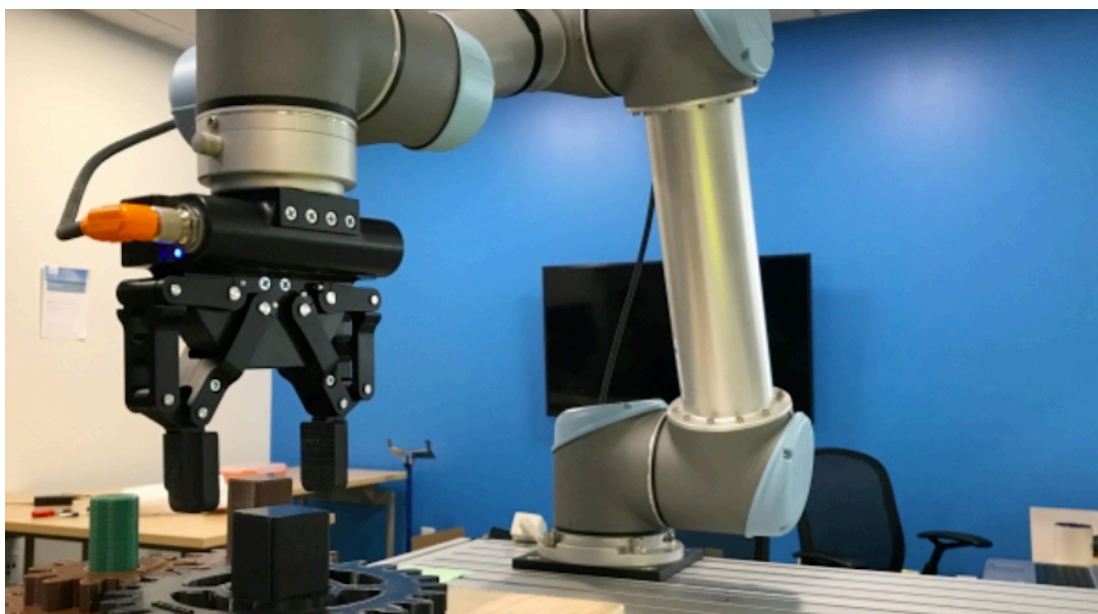


Semantic Segmentation



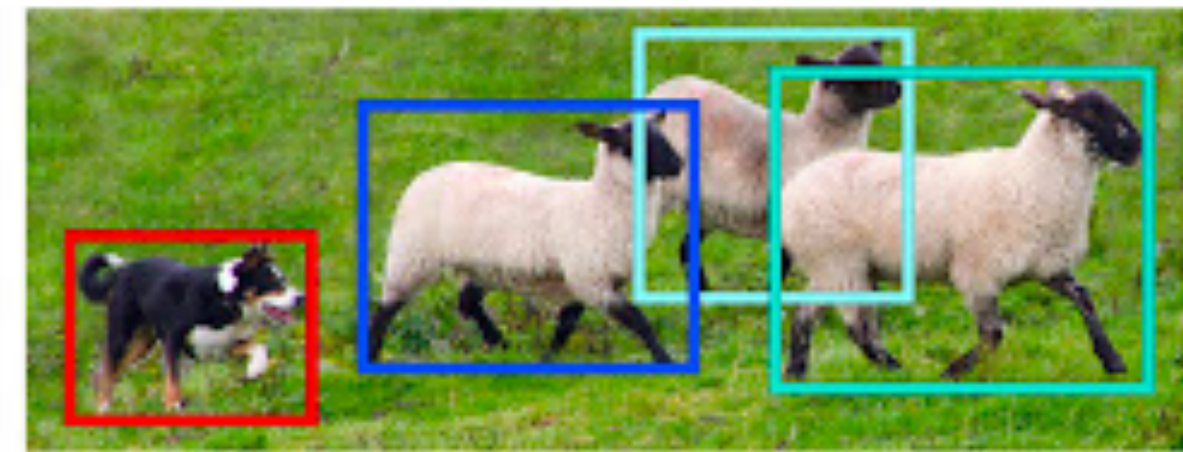
<https://medium.com/@nabilliban14/chain-colorization-using-cdcgans-and-cnn-from-learned-deep-prior-1405dab48df3>

Image colorization

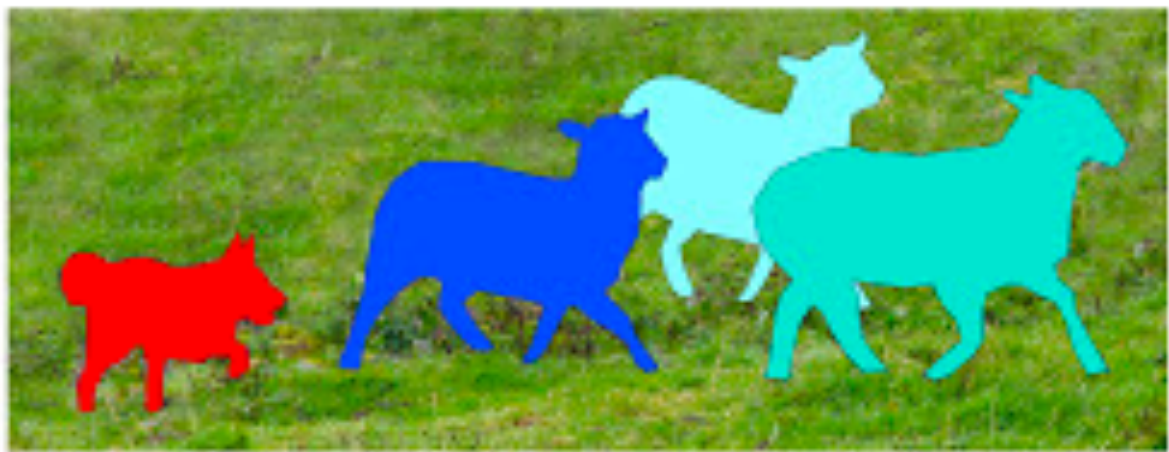


<https://www.therobotreport.com/reinforcement-learning-industrial-robotics/>

Reinforcement learning



Object Detection



Instance Segmentation

<http://manipulation.csail.mit.edu/segmentation.html>

content image



Ancient city of Persepolis

style image



The Starry Night (Van Gogh)

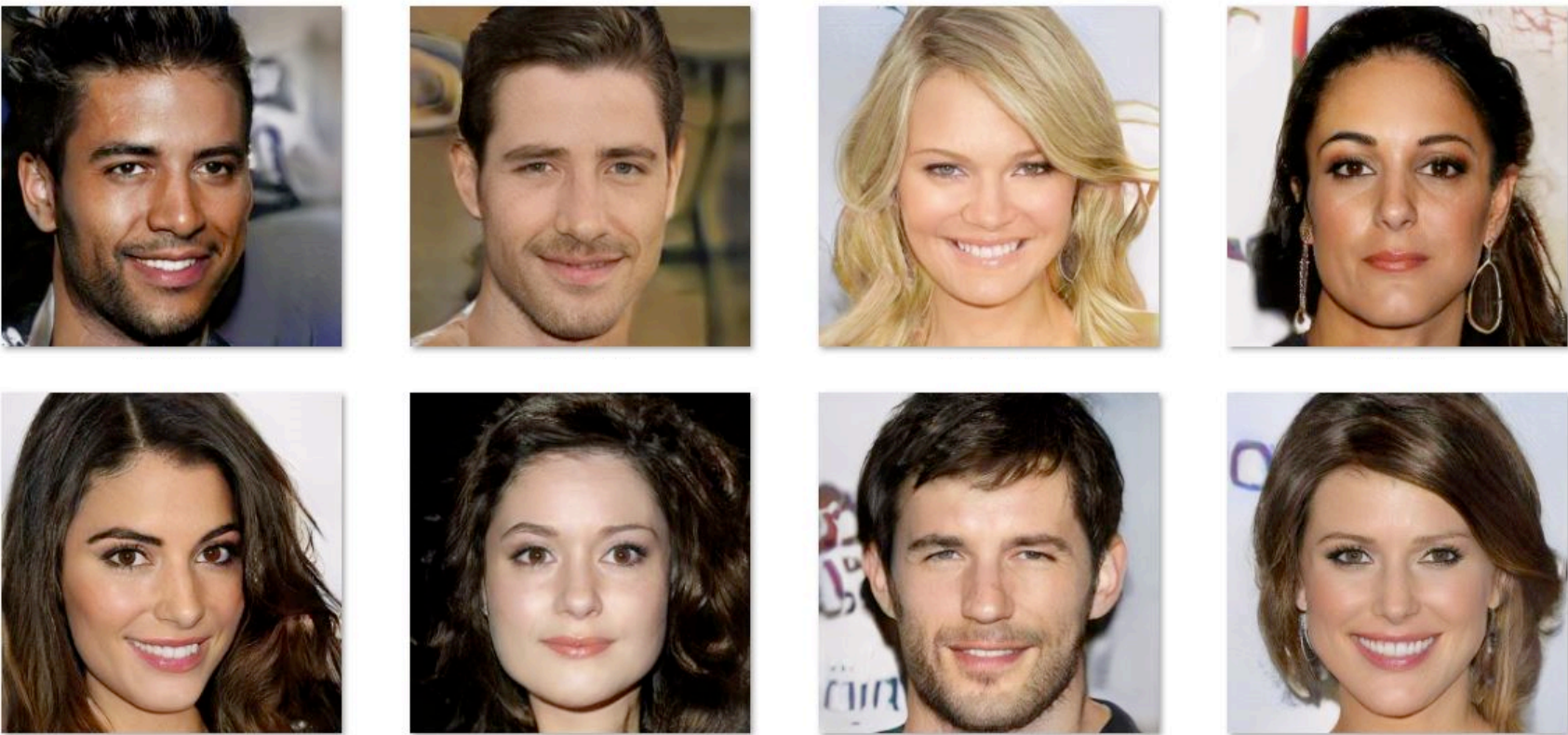
generated image



Persepolis
in Van Gogh style

<https://towardsdatascience.com/neural-style-transfer-on-real-time-video-with-full-implementable-code-ac2dbc0e9822>

Style Transfer



<https://medium.datadriveninvestor.com/artificial-intelligence-gans-can-create-fake-celebrity-faces-44fe80d419f7>

Generative models

Summary

- Image classification is a prototypical task in image processing.
- A convolution sweeps over an image, looking for a specific pattern.
- Convolutions are local (applied to image patches) and translation-invariant (applied equally to patches across the whole image).
- Convolutional neural networks are a specialized architecture for image processing, consisting of alternating convolutional and pooling layers (feature learning), following by final fully connected layer (classification).
- People have built increasingly deep CNNs, which have performed increasingly well. Image classification problem is essentially solved.
- Many other image processing tasks can be addressed with CNNs.

Hands-on: CNN in torch

- We'll now build on our previous lectures to create convolutional NNs.
- After the programming exercise, you will know how to
 - Load MNIST data (images of written digits)
 - Visualize some images
 - Build a multi-layer convolutional neural network
 - Train the CNN with stochastic gradient descent, dropout, and early stopping
 - Predict on new test images